Chimera Collapse Ages: Topology-Dependent Finite-Size Scaling in Mean-Field and Ring Oscillator Systems

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Abstract

We report a long-duration empirical study of chimera collapse in two canonical settings: the two-population Sakaguchi–Kuramoto system with strictly identical natural frequencies, on the oscillator engine of a deployed additive synthesizer at small population sizes ($\boldsymbol{N = 4}$–$\boldsymbol{64}$ per population), and the nonlocally coupled ring near its chimera existence boundary. In the mean-field system, the operational collapse detector — and the first-passage criterion standard for chimera lifetimes — measure the wrong observable: up to 98% of sustained near-synchrony crossings recover as the chimera re-forms, and 69–97% of detector firings are self-healing *grazes*. Under an absorption-grade label, true lifetimes at the operating corner ($\boldsymbol{A = 0.5}$, $\boldsymbol{\beta = 0.05}$) double, become $\boldsymbol{N}$-independent ($\boldsymbol{\tau_{abs} \approx 139}$ s over a $\mathbf{16\times}$ range of $\boldsymbol{N}$), and age (Weibull shape **2.3**–**3.0**, surviving stratification on the collective initial condition). Deaths are breath-synchronized and ratchet: per-cycle envelope maxima grow monotonically to capture. The reduced two-population model of Abrams *et al.*, validated against its published bifurcations with no fitted parameters, explains the inversion — the operating corner lies beyond the homoclinic, so the deployed "chimera" is a transient spiraling out along a destroyed limit cycle's ghost, while the nominally unreliable corner ($\boldsymbol{A = 0.2}$) holds the only true attractor — and a reduced-flow sweep shows the inversion is generic across the post-homoclinic wedge while the aging holds only beyond a finite depth, going memoryless within $\boldsymbol{\approx 0.02}$ of the homoclinic. Yet the reduced flow captures $\mathbf{3.2\times}$ faster than the finite-$\boldsymbol{N}$ system dies; direct tests rule out additive collective noise, breath-locked multiplicative noise, and the off-manifold Watanabe–Strogatz constants, localizing the prolongation to finite-$\boldsymbol{N}$ sampling on the Poisson submanifold and leaving its theory open. On the ring, just inside an independently measured existence boundary ($\boldsymbol{\beta_c = 0.137}$ $[\boldsymbol{0.130, 0.144}]$; $\boldsymbol{\beta = 0.110}$–$\boldsymbol{0.130}$, $\boldsymbol{N = 32}$–$\boldsymbol{256}$, **7,500** pre-registered runs), collapse is again an aging process: the Weibull shape exceeds one in every $\boldsymbol{(\beta, N)}$ cell and rises with $\boldsymbol{N}$ before saturating to a finite $\boldsymbol{k_\infty \approx 1.35}$–$\boldsymbol{1.65 > 1}$ (bootstrap intervals excluding one) that grows toward criticality. The dying object is a genuine chimera in **97**–**98%** of runs and the result is robust to the death criterion — but median ring lifetimes *decrease* with $\boldsymbol{N}$ on the near-boundary side, opposite to the mean-field plateau. The aging of chimera collapse is topology-independent; its finite-size scaling is topology-dependent.

Keywords: chimera states, coupled phase oscillators, Kuramoto–Sakaguchi model, nonlocal coupling, finite-size scaling, transient dynamics, Weibull aging hazard

Chimera states, in which a population of identical oscillators spontaneously splits into one synchronized group and one disordered group, are usually studied as a question in pure mathematics. Here the same phenomenon appears inside a working music synthesizer, where two coupled oscillator banks shape the sound and the software periodically judges the pattern "dead" and restarts it. Asking why it dies so often reveals that the restart trigger was watching the wrong quantity: most apparent deaths are momentary lapses that heal on their own. Once the genuine deaths are isolated, they prove regular, rhythmic, and no more frequent in large populations than small ones — and a simple reduced model explains where and why they occur, while underpredicting how long the pattern survives, a discrepancy the paper leaves as an open problem. A second, classical test bed — a ring of oscillators coupled to their near neighbors — shows the same aging of collapse risk but the opposite dependence on population size, separating what is universal about chimera death from what belongs to the network's shape.

1 Introduction

Chimera states — the coexistence of synchronized and desynchronized populations in systems of identical coupled oscillators — have been a central object of nonlinear dynamics since their discovery by Kuramoto and Battogtokh [1] and naming by Abrams and Strogatz [2]. The two-population model of Abrams, Mirollo, Strogatz, and Wiley [3] made the phenomenon solvable: two identical populations with stronger intra- than inter-population coupling support stable chimeras, breathing chimeras born in a Hopf bifurcation, and a homoclinic boundary beyond which no chimera attractor survives. At finite size the picture is enriched by transience: on nonlocally coupled rings, chimeras are chaotic transients whose lifetimes grow with oscillator count [4], and the finite-N fate of chimeras has remained a recurring theme of the literature since [5, 6], including for two small populations of the kind studied here [7].

This paper studies chimera collapse in an unusual arena: the oscillator engine of a deployed real-time additive synthesizer, in which a two-population Sakaguchi–Kuramoto bank with *strictly identical* natural frequencies drives the partial amplitudes of a musical voice, and a supervisor process re-seeds the chimera whenever an engineering detector declares it dead. Coupled-oscillator networks have an established history as sound-synthesis drivers [8, 9]; here the synthesis application matters to the dynamics only in that it fixes the regime of interest — small populations ($N = 4$–64 per side), long unattended runtimes, and a specific operating corner chosen, as it turns out, for the wrong reason. The deployment prompts a question the theory literature rarely needs to ask operationally: *what exactly is the death of a chimera, and what does a detector of it actually detect?*

The deployment fixes only the mean-field half of a larger question: which features of chimera collapse are universal, and which are properties of the coupling topology? The two-population mean field and the nonlocally coupled ring are the two canonical chimera substrates, and they make opposite finite-size predictions in their classic regimes — ring chimera lifetimes grow with oscillator count [4], while Sec. 4 shows the mean-field lifetimes are flat in N. We therefore close the paper by carrying the same survival methodology to the ring near its existence boundary (Sec. 7), where lifetimes are finite at feasible compute, asking whether the aging structure of collapse — the rising hazard — transfers across topology even where the lifetime scaling does not.

We answer with a chain of measurements, each gating the next, on bit-reproducible integrators (2,100 baseline mean-field runs plus targeted re-campaigns, and 7,500 preregistered ring runs; every figure regenerable from committed configurations). The contributions are:

1. **Graze versus absorption.** The operational collapse criterion — sustained near-synchrony of the incoherent population — overwhelmingly fires on events the chimera survives: up to 98% of criterion-satisfying crossings recover, and 69–97% of the deployed detector's firings are such grazes. We introduce a two-timescale, absorption-grade labeling and show first-passage "lifetimes" are roughly half the true absorption times (Sec. 3).
2. **An inverted stability picture.** Under absorption labeling, lifetimes at the operating corner are N-independent ($\approx 139\,\mathrm{s}$, $N = 4$–64, every seed absorbing), while at the nominally unreliable high-morph corner the majority of seeds *never* absorb: the operating corner has no chimera attractor at all, and the discarded corner holds the only persistent chimera in the system (Sec. 4).
3. **The death mechanism.** Collapse is a breath-synchronized ratchet: collapse attempts and true deaths lock to a phase of the breathing cycle, successive per-cycle envelope maxima grow monotonically (spiraling out exponentially toward the synchronization boundary), and hazard rises with cycle index. Transverse escape from the Poisson submanifold is ruled out by a validated observable and a null interventional test; lifetime is instead organized by the collective initial condition, dominantly the inter-population phase offset (Sec. 5).
4. **Closure by the reduced model — and a clean open problem.** The reduced equations of Ref. [3], validated against that paper's own published results and applied with a literal 1:1 timescale and zero fitted parameters, place the operating corner beyond the homoclinic ($A_{hc}(\beta{=}0.05) = 0.4096$), reproduce the breath period and per-cycle spiral rates, identify the never-absorbers' level with the stable chimera's r^*, and predict individual run lifetimes beyond any static feature of the seed. They also fail in one instructive way: mean-field capture is $3.2\times$ faster than finite-N death, and the flat $\tau_{\mathrm{abs}}(N)$ is absent from the N-free collective flow — finite-size fluctuations *extend* the transient (Sec. 6, Sec. 8).
5. **Topology comparison: aging transfers, scaling does not.** On the nonlocally coupled ring near its existence boundary, collapse is also an aging process: the

censored-Weibull shape exceeds one, with confidence intervals excluding one, in all 15 pre-registered (β, N) cells, and rises with N before saturating to a finite $k_\infty \approx 1.35$–$1.65 > 1$ that grows toward criticality (Appendix D). The dying object is validated as a genuine chimera in 97–98% of runs and the hazard shape is robust to the death criterion. Median ring lifetime, however, *decreases* with N in this regime — opposite to both the deep-stable ring growth [4] and the mean-field plateau: the aging of chimera collapse is topology-independent, while its finite-size scaling is topology-dependent (Sec. 7).

2 Systems

2.1 The two-population mean-field system

The voice integrates two populations $\sigma \in \{1, 2\}$ of N phase oscillators each,

$$\dot{\theta}_i^\sigma \;=\; \omega \;+\; \sum_{\sigma'} \frac{K_{\sigma\sigma'}}{N} \sum_{j=1}^{N} \sin\bigl(\theta_j^{\sigma'} - \theta_i^\sigma - \alpha\bigr), \tag{1}$$

with intra-population coupling $K_{\sigma\sigma} = \mu = (1 + A)/2$, inter-population coupling $K_{\sigma\sigma'} = \nu = (1 - A)/2$ ($\mu + \nu = 1$, $A = \mu - \nu$), and phase lag $\beta = \pi/2 - \alpha$ [10, 11], matching the conventions of Ref. [3]. A code audit confirms the natural frequencies are identical *exactly* ($\delta\omega = 0$: no per-oscillator frequency term, detuning explicitly zeroed on this path), so the dynamics is deterministic given the initial condition and collapse cannot be attributed to frequency disorder; we work in the $\omega = 0$ rotating frame. Integration is fourth-order Runge–Kutta with $dt = 0.05$ at a documented one-model-unit-per-second clock, so all times below are seconds with no conversion; halving and quartering dt leaves lifetime distributions statistically indistinguishable (two-sample KS $p = 1.0$). All runs are bit-reproducible from logged (seed, parameters, dt); every re-analysis below begins by reproducing the upstream labels exactly.

In the application, the incoherent population's per-oscillator coherences modulate the gains of an additive partial bank, $g_i' = g_i\,[\,1 + d\,a\,(c_i - \tfrac{1}{2})\,]$ with $c_i = \tfrac{1}{2}(1+\cos(\theta_i - \psi))$, turning the chimera's collective dynamics into a slowly self-morphing spectrum; the synthesis layer plays no further role in the dynamics studied here. Runs are initialized with the canonical chimera seed (one population tightly clustered, the other spread), since spontaneous chimera formation from random phases is negligible in this regime. The deployed supervisor declares the state alive while $\max_\sigma R_\sigma > 0.9$ and $\min_\sigma R_\sigma < 0.85$, fires after 2 s of sustained violation, and re-seeds the incoherent population. The baseline campaign comprises 200 seeds per population size at the operating corner ($A = 0.5$) and 100 at the high-morph corner ($A = 0.2$), $N \in \{4, 8, 16, 24, 32, 48, 64\}$, $\beta = 0.05$ throughout, integrated unsupervised to $t_{\max} = 2000\,\mathrm{s}$.

2.2 The nonlocally coupled ring

The second system is the canonical substrate of chimera-death studies, the nonlocally coupled ring [1, 4]: N phase oscillators with top-hat coupling to their P nearest neighbors on each side,

$$\dot{\theta}_k \;=\; -\frac{1}{2P+1}\sum_{|j-k|\le P}\sin\bigl(\theta_k-\theta_j+\alpha\bigr), \qquad P=\mathrm{round}(rN), \tag{2}$$

with coupling radius $r = 0.15$, phase lag $\alpha = \pi/2 - \beta$, identical frequencies in the rotating frame, and the same fixed-step RK4 integration scheme (validated against an independent reference implementation). The natural order parameter is local: $\rho_k = |(2P+1)^{-1}\sum_{|j-k|\le P} e^{i\theta_j}|$, the modulus of the windowed mean field, whose spatial profile separates a chimera's coherent arc ($\rho_k \approx 1$) from its incoherent arc (low ρ_k). Death of the ring chimera is the loss of this spatial structure: the event criterion is the spatial standard deviation $\rho_{\mathrm{std}}(t)$ of the ρ_k field falling below $\varepsilon_{\mathrm{std}} = 0.04$ and staying below for 50 time units, with right-censoring at $T_{\max} = 12{,}000$; Sec. 7 validates this criterion against alternatives and against the identity of the dying object. The ring campaign — design, event definition, primary endpoint (the censored-Weibull shape k with profile-likelihood intervals), and decision rule all pre-registered before any production run — comprises $\beta \in \{0.110, 0.115, 0.120, 0.125, 0.130\}$ approaching the existence boundary — $\beta_c = 0.137$ $[0.130, 0.144]$, measured independently in Appendix E and consistent with the literature range $\beta_c \approx 0.13$–0.14 — at $N \in \{32, 64, 128\}$ with 300 runs per cell (4,500 runs), later extended at all five β to $N = 192$ and 256 (3,000 further runs) for the thermodynamic-limit extrapolation of Appendix D.

3 Graze versus absorption

What counts as the death of a chimera? In a deployed system the answer is operational: the supervisor's aliveness condition fails continuously for 2 s, and our initial lifetime campaign used the closely related first-passage criterion that $\min(R_1, R_2)$ exceed a threshold $\theta = 0.85$ sustained for $W = 5$ s. Both definitions encode the same assumption: that a sustained excursion of the incoherent population to near-synchrony *is* the death of the chimera. This section shows that assumption is wrong, and replaces it.

3.1 Threshold crossings overwhelmingly recover

Extending the recorded trajectories past their first criterion-satisfying crossing, rather than terminating there, reveals that the crossing is usually not absorbing. At the operating corner and across $N = 8$–64, up to 98% of first crossings — excursions exceeding θ for longer than the full 5 s sustain window — are followed by recovery: the incoherent population desynchronizes again, with post-crossing minima of $\min(R_1, R_2)$ clustering at 0.2–0.45, a fully re-formed chimera. About two-thirds of these recovered trajectories are later absorbed for good; the remainder churn through further graze–recover cycles. The first sustained crossing is therefore a *graze* of the synchronization boundary, not a death, and first-passage lifetimes computed from it measure time-to-first-long-graze.

Figure 1 shows a representative trajectory ($N = 16$, $A = 0.5$; seed pinned in the released configuration). The run grazes twice — each time exceeding θ for more than 5 s before the chimera re-forms — and is absorbed on its third approach at

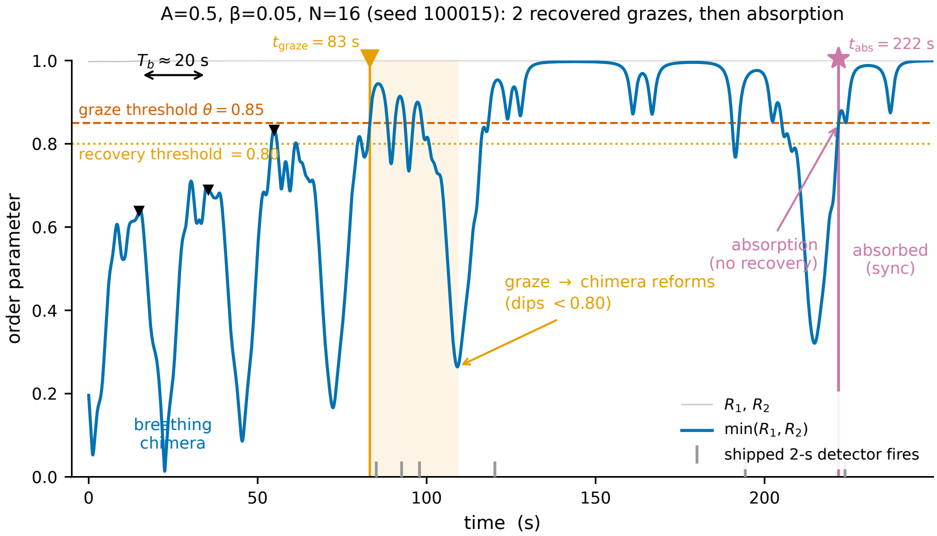

Fig. 1 A graze is not a death. $\min(R_1, R_2)$ for a single run ($N = 16$, $A = 0.5$, $\beta = 0.05$; breath period $T_b = 20.5$ s). The trajectory crosses $\theta = 0.85$ (upper line) in sustained excursions twice, recovering below the 0.80 recovery threshold (lower line) each time as the chimera re-forms, before the absorbing crossing at $t_{\text{abs}} = 221.9$ s. The engineering first-passage label $t_{\text{graze}} = 83.2$ s and the six firings of the deployed 2-s detector are marked.

$t_{\text{abs}} = 221.9$ s, having first "collapsed" by the engineering criterion at $t_{\text{graze}} = 83.2$ s. The shipped supervisor's detector fires *six times* on this trajectory before the single event that is irreversible: six firings, two reforms, one death.

3.2 An absorption-grade label

We therefore assign every run two timescales from a single trace. The first, t_{graze}, is the unmodified first-passage label, retained for comparability. The second, t_{abs}, is the time of the first crossing *not* followed by recovery — $\min(R_1, R_2) < 0.80$ sustained at least 5 s — within a verification horizon $T_v = 120$ s; runs still alternating between grazes and recoveries at the integration horizon are right-censored.

The absorption label is robust to its own knobs where it matters. Sweeping $T_v \in \{60, 120, 240\}$ s against recovery thresholds $\{0.75, 0.80\}$ moves $\hat{\tau}_{\text{abs}}$ by 22–27% at $A = 0.5$, monotonically in T_v and negligibly in the recovery threshold — comparable to or better than the $\sim$52% (θ, W)-sensitivity of the graze criterion itself. At $A = 0.2$ the apparent sensitivity is dominated by censoring rather than the labeling parameters, for a reason Sec. 4 shows is dynamical: most $A = 0.2$ trajectories never absorb at all.

The quantitative consequence at the operating corner is a factor of two: $\hat{\tau}_{\text{graze}} \approx$ 63–67 s against $\hat{\tau}_{\text{abs}} \approx 139$ s (Fig. 2). First-passage chimera lifetimes computed with engineering-style criteria here measure the wrong, and roughly halved, quantity — a gap that Sec. 7.6 shows is a property of these breathing dynamics rather than of first-passage labeling in general.

Table 1 Supervisor over-trigger: fraction of the deployed 2-s detector's firings, replayed over unsupervised traces, occurring on self-recovering grazes.

A	N	firings/run	over-trigger	median wasted (s)
0.5	8	7.2	79%	4.4
0.5	16	5.2	76%	2.4
0.5	32	4.1	73%	3.9
0.5	64	3.3	69%	5.3
0.2	8	12.3	95%	2.0
0.2	16	9.8	97%	1.8
0.2	32	7.9	96%	1.7
0.2	64	4.7	95%	1.8

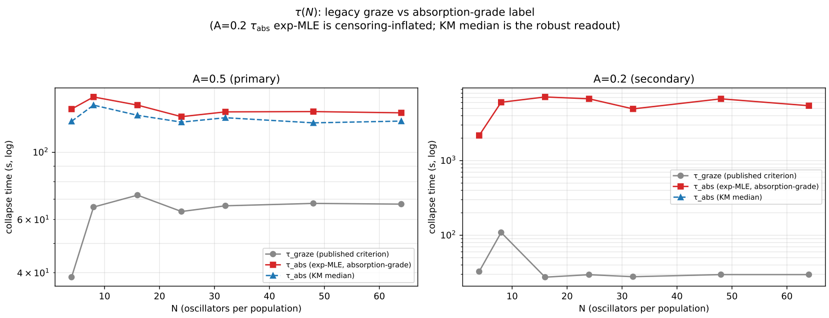

Fig. 2 Graze versus absorption lifetimes. Exponential-MLE $\hat{\tau}_{\text{graze}}$ and $\hat{\tau}_{\text{abs}}$ versus N at both coupling disparities. At $A = 0.5$ absorption lifetimes are $\approx 2.2\times$ the first-passage values and flat in N; at $A = 0.2$ no absorption lifetime is defined for the majority of seeds (Sec. 4).

3.3 The deployed detector is a graze detector

Replaying the shipped detector over the same unsupervised trajectories and classifying each firing by whether the underlying event self-recovers gives Table 1: 69–79% of firings at $A = 0.5$, and 95–97% at $A = 0.2$, occur on grazes the chimera survives on its own. The deployed system has been re-seeding a state that was mostly not dead — harmless in practice, since re-seeding lands near where the trajectory already was, but diagnostic of the conceptual gap this section closes: engineering collapse detection and dynamical absorption are different observables. All statistics below are computed on t_{abs} unless noted.

4 Absorption statistics at finite N

4.1 A flat plateau at the operating corner

At $A = 0.5$, every one of the 1,400 seeds absorbs within the horizon (0% censored at every N), and the absorption lifetime is N-independent: $\hat{\tau}_{\text{abs}} = 139, 152, 143, 131, 136, 136, 135\,\text{s}$ for $N = 4$–64, a ratio $\tau(64)/\tau(4) = 0.97$ over a sixteen-fold range of population size, with fitted exponential rate in N indistinguishable from zero. The weak rise from $N = 4$ to 16 visible in the graze times ($39 \to 72\,\text{s}$)

does not survive absorption labeling — it was a graze artifact. The published-style picture of a lifetime that grows with oscillator count, familiar from ring-topology chimeras [4], is absent here in both label and shape: adding partials does not buy persistence at this corner, at any tested N.

4.2 The inversion: the high-morph corner does not absorb

The corner the application had classified as unreliable behaves in the opposite way. At $A = 0.2$, between 51% and 78% of seeds (rising with N) *never absorb*: extending the horizon eightfold to $t_{\max} = 16{,}000\,\mathrm{s}$ leaves the censored fraction essentially unchanged, so the censoring is irreducible — a dynamical finding, not a horizon artifact. The minority that does absorb does so promptly and increasingly on the first approach: the zero-graze fraction among absorbers rises from 0.08 at $N = 4$ to 0.96 at $N = 64$. Capture at $A = 0.2$ is bimodal — absorb almost immediately, or settle into a persistent, intermittently grazing chimera [12] with no defined absorption time. The stability assumption of the deployment is therefore inverted: the "reliable" corner always dies, and the "risky" corner contains the only persistent, non-absorbing chimeras in the system. Section 6 shows both facts are exactly where the bifurcation diagram says they should be.

4.3 Aging: hazard rises with every breath

Absorption at $A = 0.5$ is not memoryless. Censored-Weibull fits give shape parameters k_{abs} of 2.27, 2.22, 2.49, 2.67, 2.44, 2.72, and 3.03 for the seven sizes $N = 4$–64, all confidence intervals well above 1 (Fig. 3) — stronger aging than in the graze times ($k_{\mathrm{graze}} = 0.82$–2.19), so the survival shoulder is not a labeling artifact.

Nor is the aging an artifact of pooling heterogeneous deterministic transients across the seed ensemble. Because the dynamics is deterministic, a fitted $k > 1$ could in principle encode only the *spread of fixed transient times* across initial conditions rather than a hazard that rises along each trajectory. We separate the two by stratifying the seed ensemble at each N into four equal-count bins of a collective initial-condition coordinate — the inter-population phase gap $|\Delta\phi_0|$, the initial incoherent-population level $\overline{R}_{\mathrm{incoh},0}$, or, decisively, the reduced model's own predicted lifetime for that seed (Sec. 6), which collapses the entire collective initial condition onto its lifetime consequence — and refitting the censored Weibull within each stratum. The aging survives in every case: $k_{\mathrm{abs}} > 1$ with 95% profile-likelihood intervals excluding one in all 84 size×stratum cells, with $k_{\mathrm{abs}} \in [2.00, 5.00]$ under the predicted-lifetime binning (likelihood-ratio $p \leq 10^{-7}$ against the exponential throughout), and the within-stratum lifetime spread remains substantial (CV = 0.22–0.56) while tracking the fitted shape through the Weibull CV–k relation at correlation 0.996 — a genuine aging-shaped spread, not a degenerate spike. Censoring at $A = 0.5$ is exactly zero (all 1,400 runs absorb before $t_{\max}$), so these are effectively uncensored maximum-likelihood fits. The aging is trajectory-level — a hazard rising along each run, as the ratchet of Sec. 5 shows directly — not initial-condition heterogeneity.

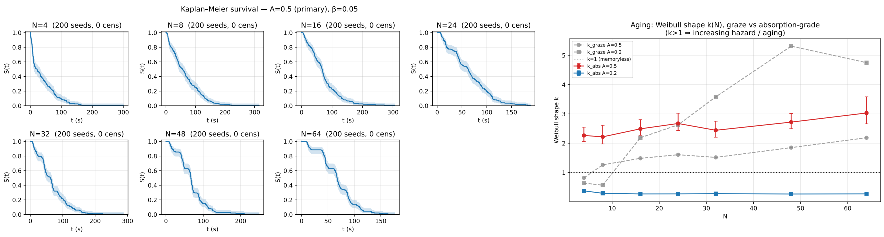

Fig. 3 Absorption survival and aging. Kaplan–Meier survival curves by N (left, $A = 0.5$) develop a shoulder with increasing N; censored-Weibull shape $k_{\mathrm{abs}}(N)$ (right) rises from 2.3 to 3.0 with all confidence intervals above the memoryless value $k = 1$.

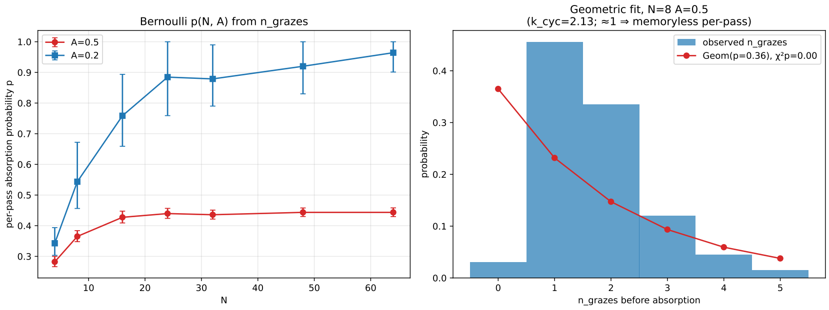

Fig. 4 Per-pass absorption statistics at $A = 0.5$. Mean per-pass absorption probability $\hat{p}(N)$ rises then plateaus, mirroring $\tau_{\mathrm{abs}}(N)$; the geometric (constant-p) model is rejected at every point — hazard rises with cycle index ($k_{\mathrm{cyc}} > 1$).

The natural per-breath Bernoulli model — a constant absorption probability per high-R pass — is rejected at every $A = 0.5$ point (χ^2 $p < 0.001$ against geometric cycles-to-absorption; Weibull-in-cycles shape $k_{\mathrm{cyc}} \approx 2.1$–$2.9$): the per-pass hazard *rises* with cycle index. The average per-pass absorption probability $\hat{p}$ nonetheless tracks the plateau, rising from 0.28 at $N = 4$ to 0.44 by $N = 24$ and flattening (Fig. 4), while the mean number of recovered grazes before death falls from 2.5 to 1.25. At $A = 0.2$ the strong apparent aging in the graze statistics (k_{graze} up to 5.3) is retired: on the few true absorptions $k_{\mathrm{abs}} \approx 0.3 < 1$, a fast, front-loaded minority rather than aging. The never-absorbers hold a mean incoherent-population level $\overline{R}_{\mathrm{incoh}} \approx 0.66$–$0.68$, sitting at the stable fixed point $r^* = 0.678$ identified in Sec. 6.

Throughout, the operative null is the memoryless one: a collapse hazard with no age dependence, $k = 1$, exponential waiting times. At the operating corner that null fails at every size and under every stratification above. It does not fail everywhere the chimera is post-homoclinic, however: sweeping the reduced flow across the (β, A) plane (Appendix C) shows the aging shape survives at the operating corner's depth beyond the homoclinic for every β tested, but within ≈0.02 of the homoclinic at $\beta \le 0.10$ the reduced collapse-time distribution is statistically indistinguishable from the memoryless null ($\hat{k} = 0.97$–1.02, intervals straddling one). Aging is a property of the corner's depth beyond the homoclinic, not of post-homoclinicity per se.

5 Mechanism: the breath-synchronized ratchet

5.1 Deaths lock to the breathing phase

The chimera at $A = 0.5$ breathes: $\min(R_1, R_2)$ oscillates with period $T_b \approx 21$–25 s, its envelope reaching ≈ 0.83 against the $\theta = 0.85$ boundary — the cycle passes within a hair of near-synchrony once per breath. Collapse *attempts* (long grazes) lock to this cycle: their phases relative to the preceding envelope peak cluster at 0.13–0.20 of the cycle, with resultant length tightening from 0.43 to 0.69 as N grows and pooled Rayleigh $p \approx 0$. True absorptions inherit the lock: pooled over $A = 0.5$, $\bar{R} = 0.21$ with Rayleigh $p = 2.2 \times 10^{-6}$ (three of six individually testable points significant; clustering weakens at large N, where capture completes within 2–3 cycles and leaves few phases to test). Mean absorption takes roughly six breaths ($\tau_{\text{abs}}/T_b \approx 6$). Death, when it comes, comes at a particular moment of the breath (Fig. 5).

5.2 The envelope ratchets monotonically to capture

Within individual runs, successive breathing maxima M_k of $\min(R_1, R_2)$ do not fluctuate about a level — they climb. The mean per-cycle increment $\langle \Delta M \rangle$ is $+0.031, +0.052, +0.078, +0.091$ for $N = 8, 16, 32, 64$, with cluster-bootstrap confidence intervals strictly positive at every point; the fraction of runs with monotone (allowing one inversion) maxima reaches 1.00 by $N = 32$. Where enough sub-threshold approach peaks exist to fit, $\log(\theta - M_k)$ falls linearly in cycle index with median slopes -0.55 to -0.77 and a negative slope in 100% of fittable runs: the approach to the synchronization boundary is an exponential spiral-out (Fig. 6). The control rules out artifacts: the same detector and windowing applied to the $A = 0.2$ never-absorbers over a median of $\sim$200 cycles gives $\langle \Delta M \rangle \approx -0.001$ and a ratchet fraction of exactly zero — stationary, with a slight *relaxation* consistent with settling onto an attractor.

5.3 What collapse is not, and what organizes it

Because the oscillators are strictly identical, a natural mechanism would be escape transverse to the Poisson (Ott–Antonsen) submanifold [13, 14] on which the reduced chimera solutions live. We tested this directly with the moment observable $D = \sum_{m=2}^{4} |Z_m - Z_1^m|^2$, validated in three stages (the Poisson identity $Z_m = Z_1^m$ to machine precision on Möbius-constructed states; dynamical invariance of on-manifold initial conditions under the shipped integrator; a clustered-state contrast), and the hypothesis fails comprehensively: initial distance D_0 does not predict lifetime ($|\rho| \leq 0.15$ at all fourteen parameter points), event-aligned $\langle D \rangle$ shows no ramp preceding collapse, and a paired interventional experiment injecting controlled D_0 levels leaves lifetimes flat. Collapse is not transverse escape.

It is instead organized in the collective coordinates. Among static features of the seed, the initial inter-population phase offset dominates: $|\Delta\phi_0|$ anticorrelates with lifetime at $\rho \approx -0.43$ to -0.59 for $N \geq 8$ (median magnitude 0.52), versus the D_0 null's ≤ 0.08, and a cross-validated regression on the $t = 0$ collective features reaches $R^2 \approx 0.2$ (quadratic up to 0.44). Larger initial phase separation between the populations means earlier death — a fact Sec. 6 converts from regression to dynamics.

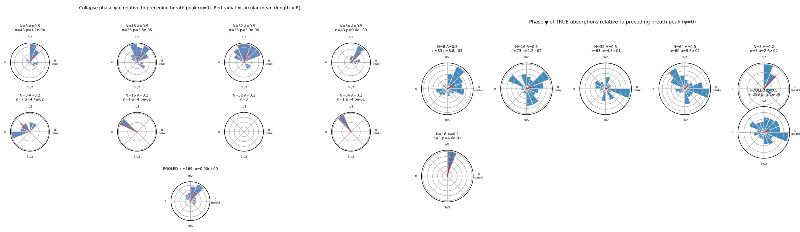

Fig. 5 Breath-phase locking. Circular histograms of collapse-attempt phases (left) and true-absorption phases (right) relative to the preceding breathing-envelope peak, $A = 0.5$. Attempts cluster at 0.13–0.20 of the cycle (pooled Rayleigh $p \approx 0$); absorptions inherit the lock (pooled $p = 2.2 \times 10^{-6}$).

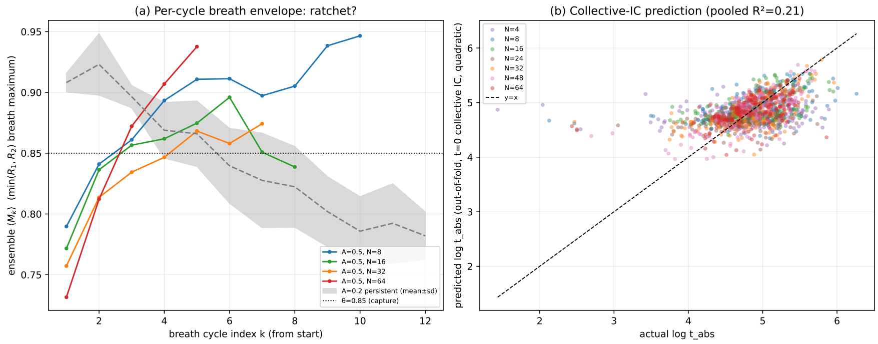

Fig. 6 The ratchet. Ensemble-averaged breathing maxima $\langle M_k \rangle$ versus cycle index k at $A = 0.5$ for several N, with the $A = 0.2$ never-absorber band (stationary, $\langle \Delta M \rangle \approx -0.001$) for contrast. Per-cycle increments are strictly positive at every N; $\log(\theta - M_k)$ falls linearly (median slopes -0.55 to -0.77).

6 Reduced dynamics: the post-homoclinic ghost

6.1 The reduced system, validated against its source

In the $N \to \infty$ mean-field, the two-population system closes in the per-population order parameters. Writing $\rho_\sigma e^{-i\phi_\sigma}$ for population σ's centroid, the reduced flow is [3]

$$\dot{\rho}_1 = -\frac{\rho_1^2 - 1}{2}\Big[\mu\rho_1 \cos\alpha + \nu\rho_2 \cos(\phi_2 - \phi_1 - \alpha)\Big], \tag{3}$$

together with the corresponding $\dot{\phi}_1$ equation and the $1 \leftrightarrow 2$ images; rotational symmetry reduces this to three variables $(\rho_1, \rho_2, \psi = \phi_1 - \phi_2)$. On the invariant manifold $\rho_1 = 1$ (synchronized population exactly locked) the flow further reduces to two variables $(r = \rho_2, \psi)$, Eq. (12) of Ref. [3], whose stationary chimeras, saddle-node curve $A_{SN}(\beta)$, and Hopf curve $A_H(\beta)$ are known in closed or series form, with the linear stability of these two-population chimeras characterized in detail by Laing [15].

Before applying any of this to our system, the implementation was gated against the source paper itself: the parametrized stationary chimeras zero the reduced right-hand side to 4×10^{-16}; integration reproduces the published regimes at $\beta = 0.1$ (stationary at $A = 0.2$; breathing at $A = 0.28$ and 0.35, with the longer period at the larger A); and numerically located saddle-node and Hopf points match the published series to 5×10^{-5}. Every number below is parameter-free and, by the integrator's documented clock, in literal seconds.

6.2 The two corners straddle the bifurcation set

At $\beta = 0.05$: $A_{SN} = 0.0948$, $A_H = 0.2703$, and the homoclinic — located by bisection on the divergence of the breathing cycle's period — at $A_{hc} = 0.4096$ (bracket width 8.8×10^{-5}). Traced over $\beta \in [0.03, 0.20]$, the homoclinic curve descends, turns, and rises into the Takens–Bogdanov point at $(0.2239, 0.3372)$ where it meets the saddle-node and Hopf curves — the non-monotone approach is demanded by the termination and serves as an internal consistency check (Fig. 7).

The verdicts follow immediately. $A = 0.2$ lies in the stable-chimera band ($A_{SN} < 0.2 < A_H$): the reduced system has a stable stationary chimera at $r^* = 0.6781$ (integration amplitude decaying to 10^{-9}), with weak relaxation $\sigma = -0.0054\,\mathrm{s}^{-1}$ — matching the never-absorbers, whose mean incoherent level sits at r^* and whose envelope drifts slightly *down*, and explaining the bimodal capture as a basin question: seeds either fall to synchrony on the first approach or onto the chimera attractor forever. $A = 0.5$ lies beyond the homoclinic ($0.5 > A_{hc} = 0.4096$): *no chimera attractor exists at the operating corner*. The chimera fixed point is an unstable spiral ($\sigma = +0.01243\,\mathrm{s}^{-1}$, $\omega = 0.3277\,\mathrm{s}^{-1}$), the breathing limit cycle that once surrounded it is destroyed, and sync is globally attracting. What the deployed voice has been operating is a transient spiraling outward along the ghost of that destroyed cycle (Fig. 8).

6.3 Quantitative match with no fitted parameters

Table 2 compares the reduced flow to the finite-N measurements. The spiral's rotation gives a linear breath period $2\pi/\omega = 19.2\,\mathrm{s}$ and a nonlinear period of $25.0\,\mathrm{s}$ for trajectories at canonical-seed amplitudes — bracketing the measured 21–25 s. The reduced per-cycle approach slopes, computed identically to the measurement from each run's seed-mapped initial condition, are $-0.62, -0.66, -0.66, -0.73$ for $N = 8$–64 against measured $-0.55, -0.50, -0.59, -0.77$: the trend with N is reproduced even though the reduced flow is N-free, because the seed ensembles at different N occupy different collective initial conditions (mean $R_{\mathrm{incoh},0}$ falls $\sim 1/\sqrt{N}$). The single substantial mismatch is the absolute capture time: the reduced median is 43 s against the measured 139 s — the mean-field flow dies $3.2\times$ faster than the finite-N system (Sec. 8). (A single trajectory started near the focus, by contrast, escapes only at $\approx 726\,\mathrm{s}$: at $\sigma = 0.0124\,\mathrm{s}^{-1}$ the $\sim$9 e-folds from a near-focus deviation take exactly that long, while the seed ensemble starts at finite amplitude — the two timescales are the same spiral read from different starting radii.)

6.4 Individual lifetimes from a three-variable model

Finally, the reduced system predicts *individual* deaths. Mapping every $A = 0.5$ seed to its collective coordinates $(\rho_1, \rho_2, \psi)_0 = (R_{\mathrm{sync}}, R_{\mathrm{incoh}}, \Delta\phi)_0$ and integrating the three-variable flow to capture yields a predicted lifetime whose Spearman correlation with

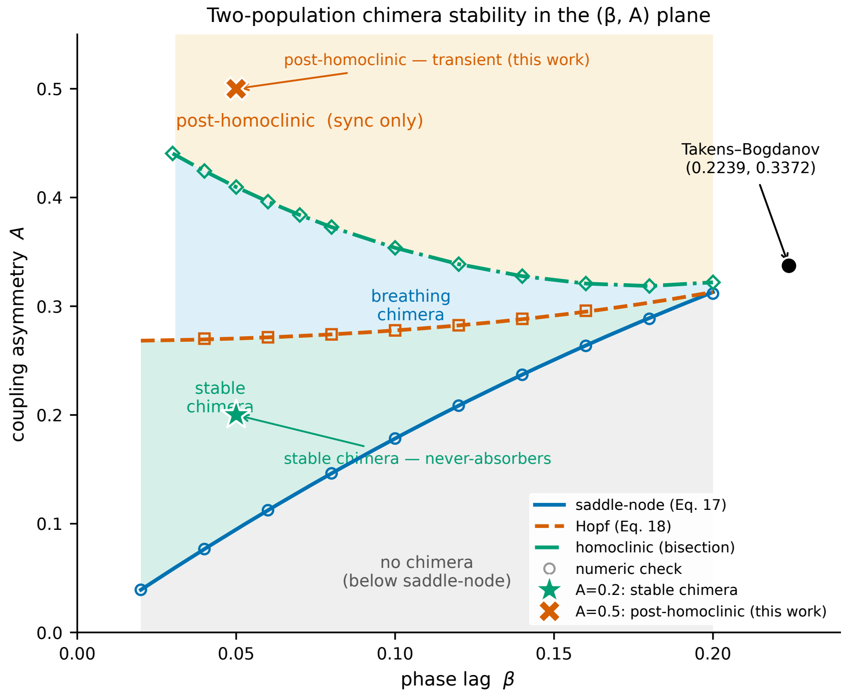

Fig. 7 Stability diagram of the two-population system in the (β, A) plane: saddle-node (Eq. 17 of Ref. [3]) and Hopf (Eq. 18) curves with numeric checks, the numerically traced homoclinic curve terminating at the Takens–Bogdanov point $(0.2239, 0.3372)$, and the two operating corners of the deployed system. The nominally stable corner ($A = 0.5$) is post-homoclinic — no chimera attractor — while the nominally unreliable corner ($A = 0.2$) sits inside the stable-chimera band.

Table 2 Reduced flow versus finite-N measurement at the operating corner ($A = 0.5$, $\beta = 0.05$). No fitted parameters; times in seconds at the integrator's documented 1:1 clock. The ratio is reduced/measured throughout; the final row reports the $A = 0.2$ chimera level for comparison.

quantity	reduced	measured	ratio (reduced/measured)
breath period T_b	25.0 (linear 19.2)	21–25	≈ 1.1
per-cycle spiral slope (pooled)	-0.67	-0.55 to -0.77	≈ 1.0
median capture time	43	139	0.31
A=0.2: chimera level	$r^* = 0.678$	$\overline{R}_{\text{incoh}} = 0.66$–$0.68$	≈ 1.0

the measured t_{abs} is 0.445 pooled (up to 0.60 per N) — comparable to the best fitted static feature — and, crucially, carries a *positive partial correlation of* +0.20–0.32 *beyond* $|\Delta\phi_0|$ *at every* N: the parameter-free dynamics predicts lifetime variance no static regression on the same inputs captures (Fig. 9). The transverse coordinate stays quiet throughout (ρ_1 deviating from 1 by ≤ 0.003 median), justifying the manifold picture a posteriori. This N-independence is not merely a mean-field artifact: for identical oscillators under global sinusoidal coupling, Watanabe–Strogatz theory [16, 17] reduces each population to a three-parameter flow at *every* N, with the remaining

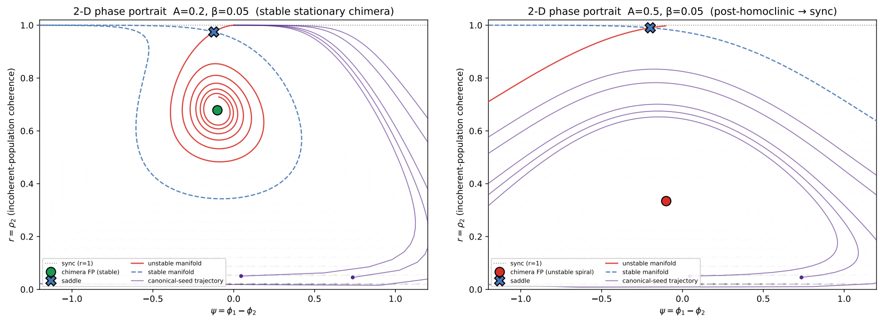

Fig. 8 Reduced-system phase portraits at $\beta = 0.05$. Left ($A = 0.2$): stable chimera spiral at $r^* = 0.678$ with the saddle and its manifolds organizing the basin. Right ($A = 0.5$): the chimera fixed point is an unstable spiral ($\sigma = +0.0124\,\mathrm{s}^{-1}$); canonical-seed trajectories spiral out to synchronization.

$N - 3$ degrees of freedom frozen as constants of motion — as the group-theoretic [18] and hierarchical-population [19] formulations make explicit. The three-variable system integrated here is exact on the uniform-constants (Poisson) submanifold, where the collective dynamics closes independently of N; the off-manifold constants, which a finite-N ensemble need not realize, are the natural suspect for the residual 3.2× prolongation — but the interventional decomposition of Appendix B exonerates them: constants-uniformized ensembles are prolonged by the same factor, so the prolongation arises from finite-N sampling of the on-manifold flow itself — sharpening, not softening, the open problem.

7 Chimera collapse on the nonlocal ring

7.1 Why the ring, and where on it

The mean-field system showed aging without lifetime growth; is either property an artifact of the all-to-all topology? The nonlocally coupled ring of Eq. (2) is the system in which finite-size chimera transience was first quantified: in the deep-stable regime its chimeras are chaotic transients whose lifetimes grow rapidly with oscillator count [4]. That regime is unmeasurably long-lived at feasible compute, so we measure the near-boundary window instead — $\beta = 0.110$–0.130, just inside the chimera existence boundary, which we measure independently at $\beta_c = 0.137$ $[0.130, 0.144]$ (Appendix E; consistent with the literature $\beta_c \approx 0.13$–0.14) — where every run dies within the horizon and survival analysis has full power. The campaign design, event criterion, primary endpoint, and decision rule were pre-registered before any production run (Sec. 2.2).

7.2 Ring collapse is a structured, aging hazard

Collapse on the ring is not a memoryless escape. Across all 15 pre-registered (β, N) cells the censored-Weibull shape $\hat{k}$ ranges from 1.13 to 1.68 with every 95% profile-likelihood interval excluding one; the likelihood-ratio test against the exponential

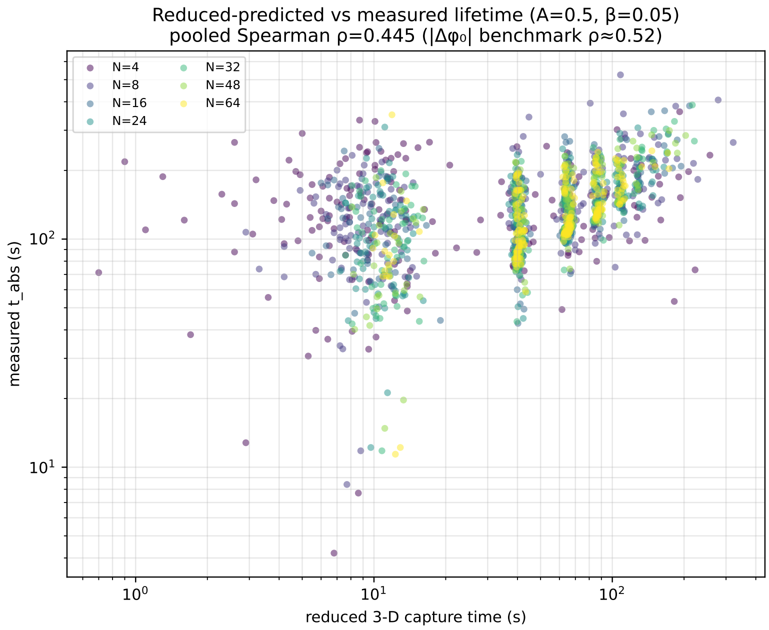

Fig. 9 Run-by-run lifetime prediction. Reduced-model capture time from each seed's collective initial condition versus measured t_{abs}, colored by N ($A = 0.5$; pooled Spearman $\rho = 0.445$; partial correlation beyond $|\Delta\phi_0|$ positive at every N).

rejects memorylessness in 15/15 cells (p from 2×10^{-2} down to 5×10^{-27}); the kernel-smoothed hazard rises visibly in every cell (max/min rise factors 1.46–2.27); and a runs test rejects linearity of $\ln \hat{S}(t)$ in 15/15 cells ($p \approx 10^{-27}$–10^{-58}). The shape is threshold-robust: re-detecting deaths on the stored traces over $\varepsilon_{\mathrm{std}} \in \{0.03, 0.04, 0.05, 0.06\}$ moves $\hat{k}$ by $\lesssim 0.07$ with the interval excluding one at every threshold. Overall censoring is 0.8% (34/4,500; worst cell 9.3% at $\beta = 0.110$, $N = 32$; exactly zero at $N \geq 64$).

The shape strengthens both toward criticality and with size: at $N = 32$, $\hat{k}$ rises from 1.13 to 1.33 across the β sweep, and at $\beta = 0.130$ from 1.33 to 1.68 across $N = 32$–128. Median lifetime meanwhile *decreases* with N — at $\beta = 0.130$ from 872 to 345 time units over $N = 32$–256 (Fig. 10a). This decreasing ring median lifetime with N at these near-boundary β reflects the near-collapse side of the existence boundary rather than the deep-stable regime of Wolfrum–Omel'chenko [4], in which lifetimes grow with N; the two regimes sit on opposite sides of the boundary and there is no contradiction between them.

Against the same memoryless null as the mean field (Sec. 4), the ring leaves no measured refuge: the exponential is rejected in every one of the 15 pre-registered cells, the refitted shape stays above one under every alternative death criterion tested below, and the null is not recovered in the thermodynamic limit — the bootstrapped k_∞ intervals exclude one at every β (Appendix D). Whether the ring possesses an analog of the mean field's near-homoclinic memoryless band (Appendix C) — a proximity to

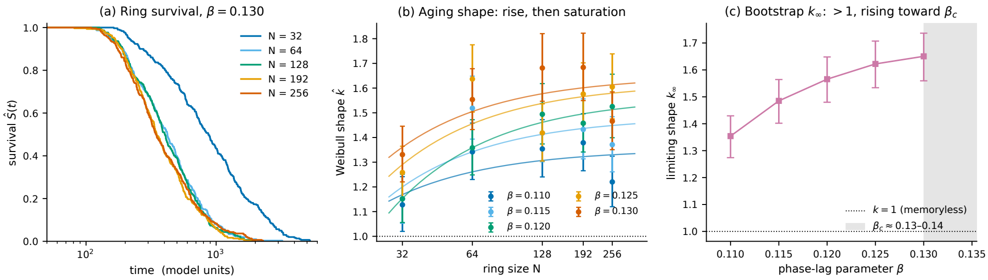

Fig. 10 Ring collapse ages, and its limiting shape is finite. (a) Kaplan–Meier survival by N at $\beta = 0.130$: median lifetime decreases with N ($872 \to 345$ over $N = 32 \to 256$) — the near-boundary side of the existence boundary, opposite to the deep-stable lifetime growth of Ref. [4]. (b) Censored-Weibull shape $\hat{k}(N)$ per β (profile-likelihood 95% intervals): all 25 (β, N) cells sit above the memoryless $k = 1$, rising with N and then saturating; curves are the AICc-selected bounded fits $k_\infty - a/N$ (Appendix D). (c) Bootstrapped k_∞ versus β: every interval excludes one, and the limiting shape rises toward the existence boundary, measured at $\beta_c = 0.137\ [0.130, 0.144]$ (Appendix E).

the boundary below which the aging shape fails — is not resolved by this campaign, whose closest approach is $\beta = 0.130$ against $\beta_c = 0.137\ [0.130, 0.144]$.

7.3 The dying object is a canonical chimera

The lifetime inversion invites the worry that the ρ_{std} criterion times a non-canonical decay channel, so we validated the dying object directly. The spatial field $\rho_k(t, \cdot)$ was not stored in the campaign, but every run is exactly reproducible from its logged seed: we re-ran all 300 trajectories per cell at $N = 256$ for $\beta = 0.110$ and $\beta = 0.130$ with field dumping, and the re-detected lifetimes reproduce the campaign values bit-for-bit (0/600 mismatches). Classifying each pre-death state by its spatial structure — an equilibration gate on the contrast (max $\rho_{\mathrm{std}} > 0.08$ over the living window, i.e. a chimera ever established) together with the dominant spatial Fourier mode of ρ_k, which is drift-invariant — shows the death criterion times the collapse of a genuine chimera, one contiguous coherent arc plus one incoherent arc or a multi-headed variant, in 98.0% ($\beta = 0.110$) and 97.3% ($\beta = 0.130$) of runs (Fig. 11). The degenerate "dissolve-then-linger" channel — the chimera disperses early and a low-contrast state persists until full homogenization — is real but rare, 2.0–2.7%. Refitting with it removed leaves the Weibull shape unchanged or slightly larger ($\hat{k} = 1.22\,[1.12, 1.32] \to 1.24\,[1.14, 1.35]$ at $\beta = 0.110$; $1.47\,[1.35, 1.58] \to 1.48\,[1.36, 1.60]$ at $\beta = 0.130$), with intervals still excluding one; sweeping the contrast gate over $[0.06, 0.12]$ only moves $\hat{k}$ further from one. The increasing-hazard claim is therefore not an artifact of the rare degenerate channel.

7.4 The hazard shape is criterion-independent

Nor is the result a property of the particular detector. Re-detecting death on the same $\beta = 0.130$ trajectories under alternative literature-style criteria — structure loss at $\rho_{\mathrm{std}} < 0.08$ and < 0.10, and a mean-coherence-level rule — the median lifetime decreases with N under every one ($\tilde{\tau}(256)/\tilde{\tau}(32) = 0.34$–$0.40$), so the inversion is a

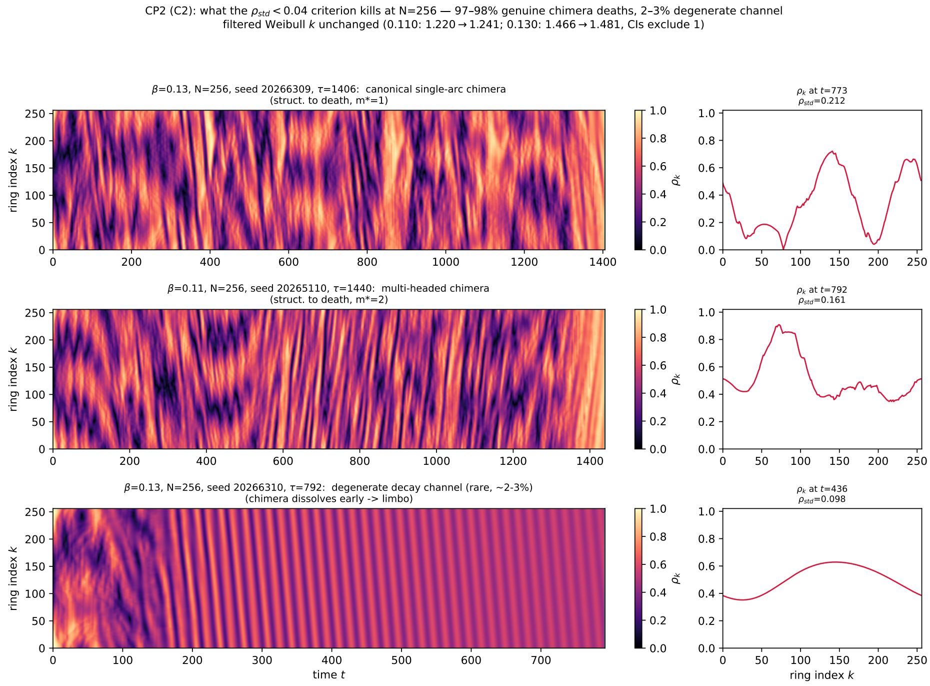

Fig. 11 What the $\rho_{\mathrm{std}} < 0.04$ criterion kills at $N = 256$. Space–time local-coherence fields $\rho_k(t, k)$ (left) with a representative mature snapshot (right) for the three pre-death behaviors: a canonical single-arc chimera (top, structured to death), a multi-headed chimera (middle), and the rare degenerate channel (bottom, the chimera disperses early and a low-contrast state lingers). 97–98% of deaths are of the first two (genuine chimera) types; removing the degenerate channel leaves the Weibull shape unchanged.

property of the near-boundary dynamics, not of the detector. The increasing-hazard shape $\hat{k}(N) > 1$ is likewise robust under the structure-loss criteria, with all intervals excluding one and 0% censoring at every N. A mean-coherence criterion (ρ-mean above a fixed level) appears to drive $\hat{k}$ below one at large N, but this is a diagnosed censoring artifact: 6.7% of the $N = 256$ collapses go to a spatially uniform *twisted* state ($\rho_k \approx 0.50$, genuinely dead by spatial homogeneity at finite time) rather than the high-coherence sync plateau, and a coherence-level threshold mislabels these genuine deaths as censored, injecting a spurious never-dying tail; re-timing them at their true deaths restores $\hat{k}(N) = 1.26, 1.44, 1.62, 1.54, 1.57$, all intervals above one (Appendix D, Fig. D6). An independent Kaplan–Meier log-cumulative-hazard slope (1.5–2.1 across N) corroborates the increasing hazard with no Weibull assumption at all. That the spatial-homogeneity criterion — unlike a coherence-level threshold — captures both the synchronized and the twisted collapse channels is itself the reason to prefer it.

7.5 Aging in the waiting time, not the descent

The structure lives in the waiting time, not in the final collapse. The terminal committed descent through the structure band is stereotyped — mean 52.8 time units, median 50.5, coefficient of variation 0.72 — and uncorrelated with lifetime (Spearman $|\rho| \leq 0.14$ in every campaign cell, ≤ 0.21 including the $N = 192$–256 extension cells) over lifetimes spanning 10^2–10^4: the longer a ring chimera has lived, the more collapse-prone it has become, while the collapse itself, once committed, takes a characteristic time independent of the age at which it occurs. This is the ring's version of the mean-field picture of Sec. 5: the hazard rises along the trajectory, and death itself is a fast, stereotyped event.

7.6 The graze/absorption gap does not transfer to the ring

Section 3 showed that in the mean-field system first-passage labels measure the wrong observable. Is that a generic defect of first-passage labeling, or a property of the mean-field dynamics? The ring campaign's stored traces cannot answer this — production runs were truncated after deep sustained collapse — so we re-ran all 300 logged seeds in three cells ($\beta = 0.110$ and 0.130 at $N = 64$, and $\beta = 0.130$ at $N = 256$) with early stopping disabled and a per-seed horizon 1,100 time units past the logged death; the re-runs reproduce the campaign lifetimes bit-for-bit (900/900). Each trajectory then received the labels of Sec. 3, pre-stated before the ensemble ran: the bare first passage t_{fp} ($\rho_{\mathrm{std}} < 0.04$, no hold), the campaign label t_{camp} (0.04 sustained 50 time units), and an absorption-grade t_{abs} — the first sub-threshold excursion that fails to recover to $\rho_{\mathrm{rec}} = 0.08$ (the equilibration-gate contrast of Sec. 7, well below the living plateau $\approx$0.15) within a recovery window $W = 50$ time units and stays below it for a verification horizon $T_v = 500$ time units, $\approx$10 times the stereotyped 52.8-unit terminal descent.

The mean-field gap does not appear (Fig. 12). The fraction of runs whose first crossing self-heals is 10.3% $[7.0, 14.0]$, 5.0% $[2.7, 7.7]$, and 3.0% $[1.3, 5.0]$ in the three cells — against up to 98% of first crossings in the mean field — and recovered excursions number 0.03–0.11 per run against 3.3–12.3 detector firings per run there. The lifetime consequence is correspondingly marginal: $\hat{\tau}_{\mathrm{abs}}/\hat{\tau}_{\mathrm{fp}} = 1.044$ $[1.022, 1.072]$, 1.028 $[1.008, 1.057]$, and 1.019 $[1.005, 1.038]$ (exponential MLE, paired bootstrap; zero censoring), where the mean field gives $\approx$2.2. A small graze population does exist — the intervals exclude zero — but it moves the measurement by 2–4%, not a factor of two. The result is insensitive to the labeling knobs: across all 27 combinations of $\rho_{\mathrm{rec}} \in \{0.06, 0.08, 0.10\}$, $W \in \{25, 50, 100\}$, and $T_v \in \{250, 500, 1000\}$, the over-trigger fraction stays within 2–20% and the ratio within 1.018–1.047. The campaign label itself is already absorption-grade: in 0/900 runs does a death labeled by the 50-unit sustain ever recover; where t_{abs} and t_{camp} differ (22–36% of runs) the absorption label is *earlier*, by at most 83 time units — descent wiggles that re-cross 0.04 without ever re-establishing contrast — moving MLE lifetimes by $\approx$1%. (Total cost: 900 re-runs, 0.18 core-hours.)

The graze/absorption distinction is therefore a property of the mean-field breathing dynamics, not of first-passage labeling per se: the two-population chimera grazes

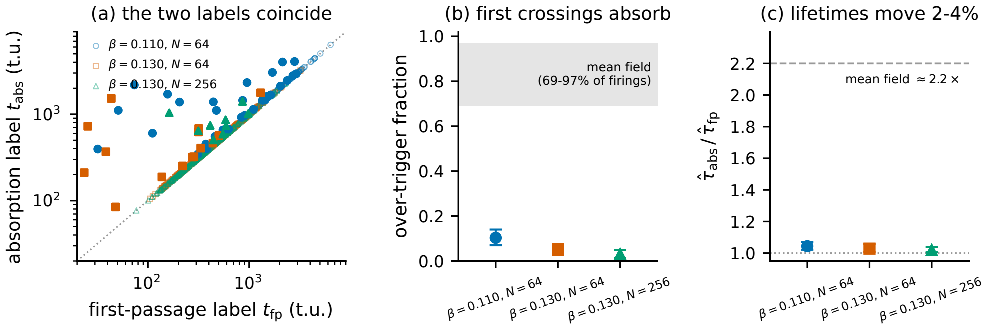

Fig. 12 The graze/absorption gap does not transfer to the ring. (a) Per-run first-passage label t_{fp} versus absorption-grade label t_{abs} for all 900 re-run campaign seeds (filled markers: the 3–10% of runs whose first sub-threshold excursion self-heals; open markers on the diagonal: first crossing = absorption). (b) Over-trigger fraction per cell (bootstrap 95% intervals) against the mean-field range of Table 1. (c) Exponential-MLE lifetime ratio $\hat{\tau}_{\text{abs}}/\hat{\tau}_{\text{fp}}$ per cell against the mean-field factor ≈2.2 of Fig. 2: absorption-grade labeling moves ring lifetimes by 2–4%.

because its breathing cycle carries the incoherent population to the synchronization boundary and back once per period, while the near-boundary ring's committed descent, once the spatial contrast falls through the death threshold, almost never re-establishes a chimera. Operationally this cuts both ways. It bounds the scope of Sec. 3's warning — first-passage lifetimes are not universally wrong, and detector design should follow the system's recovery statistics rather than a blanket rule — and it certifies the ring campaign itself: the lifetimes underlying Secs. 7–8 are absorption times, not graze times, under the same construction that doubled the mean-field numbers.

7.7 The limiting shape is finite and exceeds one

The rise of $\hat{k}$ with N poses the thermodynamic-limit question: does the shape saturate or diverge? Extending the campaign to $N = 192$ and 256 (300 runs per cell at all five β, 0% censoring) and selecting between bounded and divergent forms of $k(N)$ by AICc, the saturating form $k(N) = k_\infty - a/N$ wins in all five β cells (gaps 2.29–5.42), with bootstrapped $k_\infty = 1.35, 1.49, 1.56, 1.62, 1.65$ for $\beta = 0.110 \to 0.130$ — every interval excluding one, rising monotonically toward the measured boundary $\beta_c = 0.137$ $[0.130, 0.144]$ (Fig. 10b,c; Appendices D and E). The increasing-hazard character of ring chimera collapse is not a finite-N artifact: it survives the limit, with a limiting shape that sharpens toward criticality.

Together with Secs. 4–5, the comparison separates what is universal from what is topological. In both systems collapse ages — a hazard rising along the trajectory, expressed as a breath-locked ratchet in the mean field and as $k > 1$ at every size on the ring. The finite-size scaling, by contrast, is opposite: flat $\tau_{\text{abs}}(N)$ in the mean field, where the collective flow is N-free and N enters only through the initial-condition ensemble, versus decreasing median lifetime on the near-boundary ring (and growth in the deep-stable regime [4]), where the topology makes lifetime genuinely size-dependent. The aging belongs to the chimera; the scaling belongs to the network.

8 Discussion

The inversion, operationally. The deployed voice's celebrated reliability — $\approx 96\%$ in-state over half-hour renders — is revealed as 100% supervisor-maintained: at the operating corner there is nothing to be reliable, since every unsupervised trajectory dies in $\approx 139\,\mathrm{s}$ regardless of how many oscillators are added. Conversely the corner discarded as unreliable holds the system's only genuine attractor. For applications, two consequences follow. First, supervision is *intrinsic* to post-homoclinic operation at any N — re-seeding is not a finite-size patch but the entire persistence mechanism. Second, collapse detection should be absorption-grade: a detector with a recovery window (rather than the 2-s sustain rule, which over-triggers on 69–97% of firings) would intervene an order of magnitude less often for the same uptime.

The open problem: noise that prolongs. The reduced flow explains the geometry of death — where the corners sit, the breath period, the spiral rate, even individual lifetimes' ordering — but captures $3.2\times$ too fast, and the flat $\tau_{\mathrm{abs}}(N)$ plateau has no analog in an N-free collective flow. Both point the same direction: finite-N fluctuations *extend* the transient's life relative to mean-field, the opposite of the usual intuition in which noise destroys metastable states. At the operating corner this prolongation is N-independent — the same factor of 3.2 across the full $16\times$ range of N (against the pooled reduced median; per-N referencing shifts the $N = 64$ factor to 2.0 because the reduced ensemble's median capture crosses one additional breath cycle there) — though that independence is itself a property of the corner's depth beyond the homoclinic, not of every post-homoclinic parameter: at $\beta = 0.10$ it survives at the depth-matched point (coefficient of variation 0.14 across N) but weakens at the deeper $A = 0.5$, where the factor falls from 2.0 to 1.1 (Appendix A). A natural candidate for the prolongation is fluctuation-induced re-injection — finite-size kicks scattering the trajectory back toward the ghost region the deterministic spiral is leaving. We tested the simplest version directly (Appendix B): additive Gaussian noise of amplitude $c/\sqrt{N}$ on the collective variables of the reduced flow. It does produce re-injection and prolongation, but the prolongation is N-*dependent* — strong at small N and absent by $N = 64$ — because the noise amplitude is. The observed N-independent factor of 3.2 at the operating corner therefore rules out additive collective noise. The two candidates this leaves — multiplicative fluctuations tied to the breath cycle, and the off-manifold Watanabe–Strogatz constants of Sec. 6 — are both ruled out in Appendix B: breath-locked multiplicative noise at its measured amplitude leaves capture essentially unperturbed and at no amplitude produces more than a $1.4\times$ median prolongation (envelope-gated noise vanishes exactly where re-injection must act), while seeds whose constants are projected onto the uniform submanifold at machine precision remain prolonged by the same factor as the real seeds. What survives is sharper than a candidate list: the prolongation is generated by finite-N sampling of the collective flow *on* the Poisson submanifold, and it accrues mainly between the first sustained θ-crossing and absorption — the fluctuation-induced recoveries the deterministic reduced flow does not possess. We report the mechanism's location as measured and leave its theory open; the rising-hazard (aging-in-cycles) structure and the breath-locked phase distribution remain sharp constraints any such theory must satisfy.

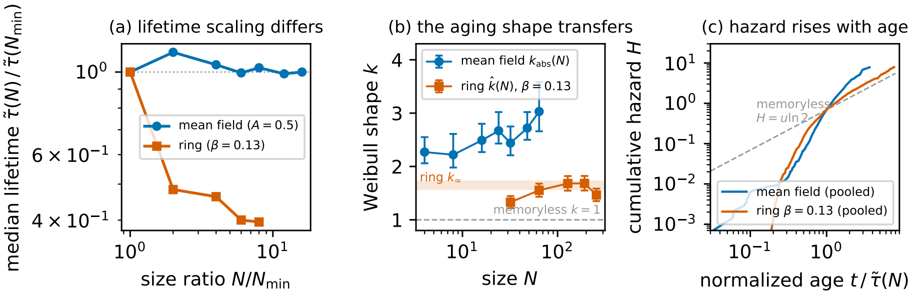

Fig. 13 Cross-system comparison: aging transfers, scaling does not. (a) Median lifetime versus size, each system normalized to its smallest campaign size: the mean-field absorption plateau is flat over a 16× range of N ($\tilde{\tau}(64)/\tilde{\tau}(4) = 1.00$), while the ring's near-boundary median falls 2.5× over an 8× range (872 → 345 time units at $\beta = 0.130$). (b) Censored-Weibull shape versus size for the same campaigns: mean-field $k_{\mathrm{abs}} = 2.27$–3.03 and ring $\hat{k} = 1.33$–1.68, every interval above the memoryless $k = 1$ (dashed); the band is the ring's bootstrapped thermodynamic limit $k_\infty = 1.65$ $[1.56, 1.74]$ at $\beta = 0.130$ (Appendix D). (c) Nelson–Aalen cumulative hazard versus normalized age $t/\tilde{\tau}(N)$ (lifetimes median-normalized per cell, pooled within each system; zero censoring in both campaigns): both curves are convex against the memoryless reference $H = u \ln 2$ — the hazard rises with age in both systems — with the ring's late bend back toward slope one marking its front-loaded rise.

Topology: aging transfers, scaling does not. On nonlocally coupled rings in the deep-stable regime, finite-size chimeras are chaotic transients with lifetimes growing rapidly in N [4]. The two-population mean-field case is qualitatively different: its collective flow is N-free, the N-dependence entering only through the initial-condition ensemble — and correspondingly we find no lifetime growth at all. Section 7 turns this contrast from a citation into a measurement (Fig. 13). On the near-boundary side of the ring's existence boundary, median lifetime *decreases* with N, under every death criterion tested, while the hazard keeps the same aging signature as the mean field: $\hat{k} > 1$ in every cell, rising with N and saturating to $k_\infty \approx 1.35$–$1.65 > 1$ (bootstrap intervals excluding one) in the thermodynamic limit. In both systems the aging is trajectory-level — surviving conditioning on the collective initial condition in the mean field (Sec. 4), and robust to the death criterion and to the identity of the dying object on the ring. The N-scaling of chimera transience is a property of the coupling topology; the aging of the collapse hazard is not.

What unifies, and at what level. The comparison supports one mechanism-level statement, provided it is made at the level of the hazard. In both systems the collapse hazard rises along the trajectory: model-free, the death rate after the median age exceeds the rate before it by factors of 3.5–5.1 across the mean-field sizes and 1.4–2.2 across the $\beta = 0.130$ ring cells (Fig. 13c) — though the two systems expose that rising hazard very differently. In the mean field it has a measured monotone carrier: the per-cycle envelope ratchet of Sec. 5, climbing the unstable spiral that the reduced flow guarantees everywhere in the post-homoclinic wedge ($\sigma > 0$ at all 592 grid points; Appendix C). On the ring we find no such observable, and not for lack of looking: the terminal descent is stereotyped and age-uncorrelated (Sec. 7), and the one per-run statistic that does track lifetime — the structure level averaged over the run's life, Spearman $\rho = 0.69$–0.81 — does so mechanically, because its averaging window contains the terminal descent, which drags down short runs' averages; restricted to runs beyond twice the median lifetime, where the descent occupies at most $\sim$10% of the window, the correlation collapses to $|\rho| \leq 0.23$ (all $p > 0.13$). Aging as a monotone *observable* approach to the absorbing boundary is therefore mean-field-specific; what transfers across topology is only the rising hazard. Nor is a monotone approach sufficient even where it exists: $\sigma > 0$ holds at every post-homoclinic point, yet within $\approx$0.02 of the homoclinic at $\beta \leq 0.10$ the capture-time distribution is memoryless — two of the three failing points put 12–28% of captures beyond five median lifetimes (no aging point exceeds 2%) and the third sits at exponential-level relative dispersion, the long ghost-cycle passages near the saddle dispersing the waiting times the drift would otherwise order (Appendix C). Where collapse ages, then, it ages because monotone drift toward absorption dominates the dispersal of passage times; where that drift surfaces in a collective observable, deaths ratchet. The scaling half never transfers, and for a stateable reason: it is set by how N enters the collective flow — not at all in the mean field (the flat plateau, $\leq$14% variation over a 16$\times$ range of N), structurally on the ring, where even the sign is boundary-side dependent (the near-boundary median falls 2.5$\times$ while the deep-stable regime grows [4]). The aging belongs to the trajectory's drift toward the absorbing state; the scaling belongs to how the network enters the flow.

Scope. All results are at strictly identical frequencies, equal population sizes, the shipped phase lag ($\beta = 0.05$, with the phenomenology reproduced at $\beta = 0.10$; Appendix A), and one canonical seed family; the absolute τ_{abs} carries the documented 22–27% labeling sensitivity; and the collective-coordinate summaries are a lossy projection of the $2N$-phase state (weakest at $N = 4$, where they predict little). Within that scope, every headline claim is criterion-robust, bit-reproducible, and regenerable from committed configurations.

9 Conclusion

An engineering question — why does a synthesizer's chimera need so much re-seeding? — unwound into a dynamical one. The collapse detector turns out to detect grazes, not deaths: most sustained near-synchrony excursions heal, and true absorption takes twice as long as first passage suggests. Corrected lifetimes are N-independent and aging, deaths lock to the breathing phase and arrive by a monotone ratchet of the envelope, and the bifurcation structure of the reduced two-population model explains why: the operating corner lies beyond the homoclinic, so the deployed state is a transient on a destroyed cycle's ghost, while the discarded corner holds the true chimera attractor. A three-variable, parameter-free flow predicts the period, the spiral rate, and the ordering of individual deaths — and underestimates their timing by 3.2$\times$, leaving one clean open problem: at finite N, fluctuations make this chimera live longer, not die faster.

Carrying the same survival methodology to the canonical ring substrate then shows which half of the story is universal. Near its existence boundary the ring's chimera also dies by an aging process — Weibull shape above one in every pre-registered cell, saturating to $k_\infty \approx 1.35$–$1.65 > 1$ in the thermodynamic limit, with the dying object validated as a genuine chimera and the shape robust to the death criterion — while its lifetimes *fall* with N, opposite to the mean-field plateau and to the deep-stable ring growth. Chimera collapse ages regardless of topology; how long the chimera lives, and how that lifetime scales with size, belongs to the network.

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Declarations

Funding

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Competing interests

The author declares no competing interests.

Data and code availability

The data and code that support the findings of this study — the integrators, experiment harnesses, campaign records, committed configurations, and figure-generation scripts — are openly available in the annealMusic repository at github.com/akarlin3/annealMusic and archived at Zenodo (doi.org/10.5281/zenodo.20564867); `python3 tools/paper-figures/run_all.py` regenerates the mean-field figures, the ring summary figure (`fig10_ring.py`), the cross-system comparison (`paper_figures/fig12_crosssystem.py`, hazard-level checks in `tools/unify_checks.py`), and the measurements added in revision: the corner-generality map (`paper_figures/fig_corner_map.py`, data from `tools/reduced-ode/corner_map_data.py`), the re-injection mechanism probe (`paper_figures/fig_mech_probe.py`, data from `tools/noise-test/mech_probe.py` and `tools/manifold-probe/ws_decomposition.mjs`), the existence-boundary measurement (`paper_figures/fig_betac.py`, data from `anneal-hazard/analysis/run_betac.py`), the depth-onset reanalysis (`paper_figures/fig_depth_aging.py`, data from `tools/reduced-ode/depth_aging_fit.py`), and the ring graze/absorption labeling measurement (`paper_figures/fig_xlabel.py`, data from `tools/xlabel/run_xlabel.py`). The ring campaign (pre-registration, ensemble records, and fits) is in `anneal-hazard/`; the stratified-hazard and criterion-robustness analyses, including the field re-runs validating the dying ring object, are in `critical_review/` and were independently reproduced with a disjoint censored-Weibull implementation.

Author contributions

Avery Karlin: Conceptualization (lead); Methodology (lead); Software (lead); Formal analysis (lead); Investigation (lead); Data curation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead).

Appendix A Robustness of the phenomenology at $\beta = 0.10$

To show the operating-corner phenomenology is not an artifact of the shipped phase lag $\beta = 0.05$, we repeated the finite-N campaign at $\beta = 0.10$, the value at which the reduced model is validated against the published regimes [3]. Reduced-model placement gives a saddle-node at $A_{SN} = 0.178$, a Hopf at $A_H = 0.278$, and a homoclinic at $A_{hc}(0.10) = 0.354$ (bisection bracket width 9×10^{-5}). We sampled the post-homoclinic corner $A = 0.5$ (parallel to the operating corner) and, to match its distance beyond the homoclinic, a depth-matched point $A = A_{hc} + 0.090 = 0.444$; $A = 0.2$ and a cleaner mid-band point $A = 0.24$ sit in the stable-chimera band as never-absorber controls.

Every headline result survives (Fig. A1). The absorption lifetime is flat in N ($\hat{\tau}_{\text{abs}}(64)/\hat{\tau}_{\text{abs}}(8) = 0.93$ at $A = 0.5$; 0.87 at $A = 0.444$; fitted exponential rate $\approx -1 \times 10^{-3}$ per oscillator, zero censoring). The censored-Weibull shape stays above one ($k_{\text{abs}} = 1.8$–2.8 at $A = 0.5$, 1.4–1.6 at $A = 0.444$), so hazard still rises. The per-cycle ratchet is positive ($\langle \Delta M \rangle = +0.06$ to $+0.19$, bootstrap CIs excluding zero), and true absorptions remain breath-phase-locked (pooled Rayleigh $p = 6.5 \times 10^{-5}$). The reduced flow reproduces the breath period ($\approx$ 23–25 s, measured 20–25 s) and spiral rate. The stable-band controls behave as expected: the mid-band $A = 0.24$ holds a near-N-independent persistent fraction (44% at $N = 8$, 52% at $N = 64$; 50 seeds each), a cleaner never-absorber control than $A = 0.2$, whose persistent fraction swings more widely (28% to 60%) as it sits close to A_{SN}.

One caveat refines a body-text claim. The transient is shallower at $\beta = 0.10$ (≈ 2 breath cycles to absorption vs ≈ 6 at the operating corner), and correspondingly the reduced-to-measured capture ratio — the 3.2× prolongation of Sec. 8 — is N-independent only at comparable post-homoclinic depth. At the depth-matched $A = 0.444$ it stays flat (coefficient of variation 0.14 across N), but at the deeper $A = 0.5$ it falls from 2.0 to 1.1 across N. The N-independence of the prolongation is thus a property of the operating corner's depth beyond the homoclinic, not of every post-homoclinic parameter.

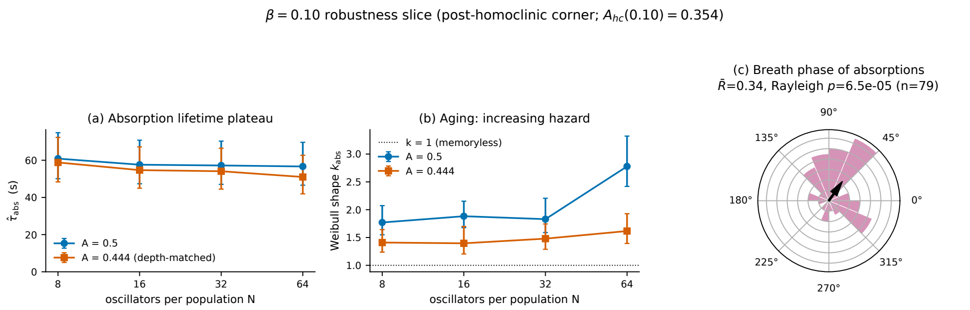

Fig. A1 Robustness at $\beta = 0.10$. (a) Absorption lifetime $\hat{\tau}_{\text{abs}}(N)$ is flat in N at both the post-homoclinic corner $A = 0.5$ and the depth-matched $A = 0.444$. (b) Censored-Weibull shape $k_{\text{abs}}(N)$ stays above the memoryless value $k = 1$ (aging persists). (c) True-absorption breath phases remain clustered (pooled $\bar{R} = 0.34$, Rayleigh $p = 6.5 \times 10^{-5}$, $n = 79$).

Appendix B A finite-size noise test of fluctuation re-injection

The reduced flow captures $\approx 3.2\times$ faster than the finite-N system, and at the operating corner that prolongation is N-independent (Sec. 8). The natural candidate — fluctuation-induced re-injection — predicts that adding finite-size noise to the collective flow should reproduce both facts. We tested this directly, integrating the three-variable reduced flow at the operating corner ($A = 0.5$, $\beta = 0.05$) with the Euler–Maruyama scheme and adding Gaussian noise of amplitude $\sigma_{\text{noise}} = c/\sqrt{N}$ to the collective variables (ρ_1, ρ_2, ψ), with reflection keeping ρ in $[0, 1]$. At $c = 0$ the scheme reproduces the deterministic capture times to a median 0.25% per initial condition. The physical scale, estimated from the measured high-frequency fluctuation of R_{incoh} in the baseline data, is $c \approx 0.05$. We swept $N \in \{8, 16, 32, 64\}$, $c \in \{0, 0.025, 0.05, 0.1, 0.2\}$, 200 realizations each, from the Sec. 6 seed-mapped initial conditions.

The result is a clean negative (Fig. B2). At the physical scale $c \approx 0.05$ the noise barely perturbs capture (prolongation factor ≈ 0.97, N-independent). Larger amplitudes do prolong — at $c = 0.2$ the trajectory is repeatedly re-injected, accumulating up to $\sim$240 breath cycles before capture — but the prolongation is strongly N-*dependent* (factor 2.4 at $N = 8$ falling to 0.7 at $N = 64$; coefficient of variation 0.40 across N), and never reaches the measured ≈ 3.2 at any cell where it is N-independent. The rising-hazard constraint holds only at small c ($k > 1$ for $c \leq 0.05$, $k \approx 1$ for $c \geq 0.1$), while breath-phase locking survives throughout. Thus additive $1/\sqrt{N}$ noise on the collective flow reproduces the mechanism (re-injection) but with the wrong N-scaling: because its amplitude is N-dependent, so is the prolongation it produces. The N-independent $3.2\times$ factor requires a different mechanism.

The two candidates this negative left standing — multiplicative fluctuations tied to the breath cycle, and the off-manifold Watanabe–Strogatz constants of Sec. 6 — were then tested directly. Both fail, each in an instructive way.

Breath-locked multiplicative noise. We repeated the Euler–Maruyama experiment with the noise amplitude tied to the breath envelope and carrying no N factor at all:

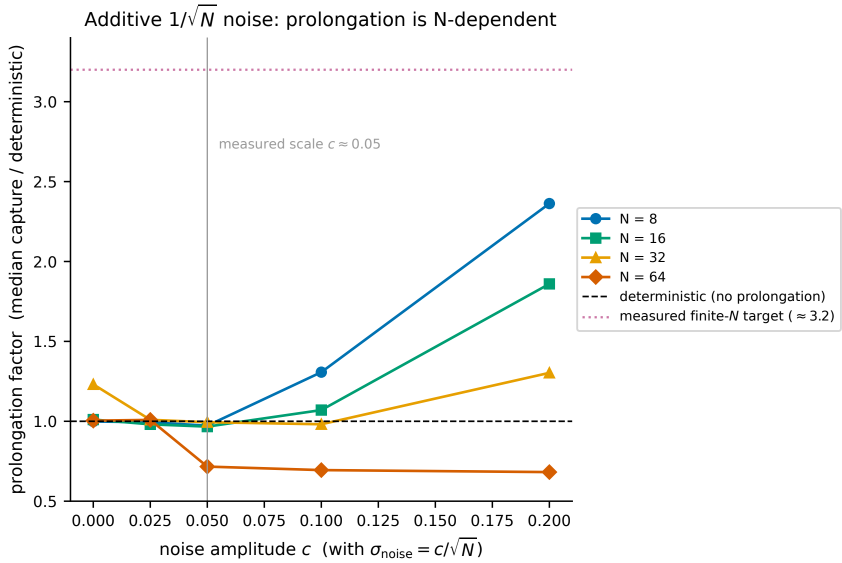

Fig. B2 Finite-size noise test. Prolongation factor (median capture relative to the deterministic reduced flow) versus additive-noise amplitude c ($\sigma_{\mathrm{noise}} = c/\sqrt{N}$) at $A = 0.5$, $\beta = 0.05$. At the physical scale $c \approx 0.05$ the curves sit near unity; at larger c they fan out with N, so the prolongation is N-dependent and never reaches the measured ≈ 3.2 (dotted) where it is N-independent.

$\sigma_{\mathrm{eff}} = c_m\,[1 - \min(\rho_1, \rho_2)]$, strongest mid-breath and vanishing at capture, applied to (ρ_1, ρ_2, ψ) with the same reflection ($c_m = 0$ reproduces the deterministic capture to a median 0.23–0.25% per initial condition). The physical amplitude, from the same baseline traces as Appendix B's $c \approx 0.05$, is the high-frequency R_{incoh} fluctuation relative to the envelope: $c_m = \sigma_{\mathrm{HF}}/\langle 1 - R_{\mathrm{incoh}}\rangle = 0.052$, 0.040, 0.028, 0.022 for $N = 8$–64 (pooled median 0.034) — already N-dependent, which by itself undercuts the premise of an N-free multiplicative amplitude. The sweep ($c_m = 0.5\times$–$16\times$ the pooled estimate, 200 realizations per cell, no censoring) is a clean negative (Fig. B3a): at the physical amplitude the prolongation factor is 0.93–1.00 for $N = 8$–32 (0.69 at $N = 64$), and at *no* amplitude does the median factor exceed 1.39 — strong envelope noise eventually *accelerates* capture (factor 0.52 at $16\times$) rather than prolonging it. The reason is structural: noise gated by the breath envelope is weakest exactly where re-injection must act, near the capture boundary ($\sigma_{\mathrm{eff}} \le 0.15\,c_m$ above θ). Breath-phase locking and rising hazard survive at the physical amplitude (Rayleigh $p < 10^{-4}$ at every N; $k_{\mathrm{cyc}} = 2.6$–3.4 with 95% profile intervals excluding 1), but the prolongation conditions fail decisively.

Off-manifold Watanabe–Strogatz constants. For identical oscillators the finite-N dynamics is the WS flow with the seed's constants, so the constants hypothesis admits an interventional test with the campaign integrator itself. For every $A = 0.5$ campaign seed at $N \in \{8, 16, 32, 64\}$ (200 seeds per N) we built a *constants-uniformized partner*: per population, a Möbius push of the exact splay configuration matched to the seed's complex Z_1 (match error $\le 9\times10^{-16}$), i.e. the same collective state with the constants

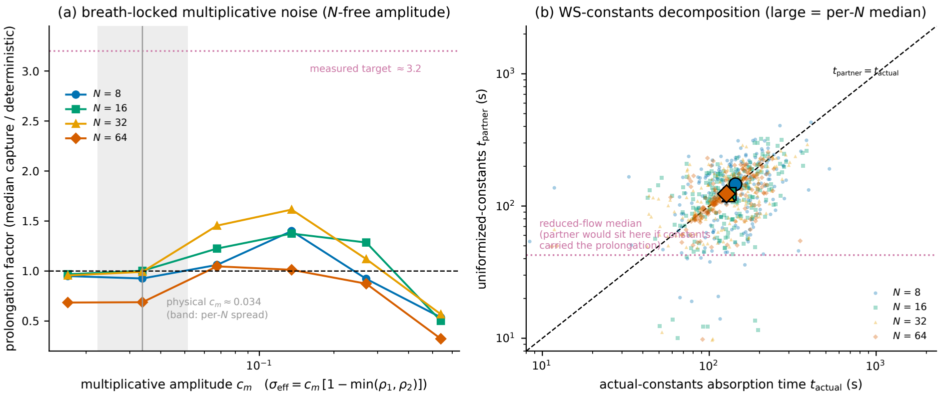

Fig. B3 The two remaining re-injection candidates, tested. (a) Multiplicative breath-locked noise $\sigma_{\rm eff} = c_m[1 - \min(\rho_1, \rho_2)]$ (N-free amplitude by construction): prolongation factor versus c_m at $A = 0.5$, $\beta = 0.05$. At the physical amplitude (vertical line; band = per-N spread of the measured estimate) the factor is ≈ 1, and no amplitude raises it beyond 1.4 — envelope-gated noise cannot re-inject from the capture region, and at large c_m it accelerates capture. (b) Constants decomposition: absorption time of the constants-uniformized partner versus the actual seed (200 seeds per N; large symbols are per-N medians). The pairs sit on the diagonal, a factor ≈ 3 above the reduced-flow median (dotted): uniformizing the Watanabe–Strogatz constants does not remove the prolongation.

projected onto the uniform (Poisson) submanifold — the incoherent population's closure defect drops from a median $D_0 = 0.04$–0.35 to its splay floor (down to 10^{-31}). Actual and partner were integrated with the identical absorption-grade pipeline (the actual runs reproduce the recorded $t_{\rm abs}$ exactly). If the off-manifold constants carried the prolongation, the partners would die on the reduced flow's schedule (≈ 43 s; 64 s at $N = 64$). They do not (Fig. B3b): partner medians are 146, 124, 120, 124 s against actual 143, 133, 130, 127 s — actual-to-partner ratios 0.98–1.08, bootstrap intervals all spanning 1 — so ensembles with *machine-precision uniform constants* are prolonged by the same $\approx 3\times$ factor as the real seeds. Consistent with the D_0 null of Sec. 5, the constants are exonerated interventionally, not just correlationally. The prolongation must therefore be generated by the finite-N sampling dynamics *on* the Poisson submanifold; the decomposition also localizes it within the run — the first sustained θ-crossing comes only 1.0–1.6$\times$ later than the reduced capture, and the bulk of the factor accrues between that first graze and absorption (a mean 1.2–1.7 fluctuation-induced recoveries per run), exactly the window in which the reduced flow, which never recovers, has already died.

Appendix C Generality of the post-homoclinic regime across the (β, A) plane

Section 6 and Appendix A establish the post-homoclinic inversion at two phase lags. This appendix asks, within the reduced model, whether the inversion — and the aging shape of the collapse-time distribution — is a property of those corners or of the post-homoclinic region as a whole. We traced the homoclinic curve $A_{hc}(\beta)$ at 35 values of $\beta \in [0.03, 0.20]$ by the escape-to-sync bisection of Sec. 6 (bracket width $\leq 7 \times 10^{-5}$

everywhere; $A_{hc}(0.05) = 0.4096$ reproduced exactly), combined it with the saddle-node and Hopf series (Eqs. 17–18 of Ref. [3]; numeric locators agree to 1.4×10^{-3} and 8.7×10^{-4}), and classified a 40×40 grid over $(\beta, A) \in [0.03, 0.20] \times [0.02, 0.55]$ into the four regimes of Fig. 7 (Fig. C5).

The inversion is generic. At every one of the 592 grid points above the homoclinic curve the stationary-chimera fixed point of Eqs. (13)–(14) persists and is an unstable spiral: $\sigma = \mathrm{Re}\,\lambda > 0$ at all 592 points, with $\sigma = 0.0046$–$0.0578\,\mathrm{s}^{-1}$ and $\omega = 0.14$–$0.35\,\mathrm{s}^{-1}$, σ growing with β and with depth beyond the homoclinic (Fig. C5); the operating corner reproduces the Sec. 6 values ($\sigma = +0.01243\,\mathrm{s}^{-1}$, $\omega = 0.3277\,\mathrm{s}^{-1}$). Sync is globally attracting in practice as well as in the portrait: every ensemble run below captured (zero censoring in 9,600 integrations), as did single canonical-seed trajectories at three additional spot points.

The aging shape is *not* uniform over this region. We selected eight post-homoclinic points spanning the wedge — four shallow points at fixed depth $A - A_{hc} \approx +0.020$ with $\beta = 0.03$–0.18, and four deep points at $A = 0.5$ with $\beta = 0.03$–0.16, including the operating corner — and at each integrated the full Sec. 6 canonical-seed ensemble (the 1,200 seed-mapped collective initial conditions at $N = 8$–64, 200 per size) through the three-variable flow to capture, $t_{\max} = 2000\,\mathrm{s}$. (The 200 $N = 4$ rows are excluded: ten of them start beyond the capture threshold — the two-oscillator "population" makes the seed map degenerate, producing $t = 0$ events on which the Weibull likelihood is undefined; including them via a clamped fit moves the corner value from $\hat{k} = 1.40$ to 1.22 and changes no verdict.) The censored-Weibull shape was fitted with the profile-likelihood MLE of Sec. 4 and, independently, with a Nelder–Mead MLE with 1,000-resample bootstrap intervals; the two point estimates agree to better than 10^{-5} at every point (Table C1).

Aging survives everywhere at the operating corner's depth, and fails in a narrow band along the homoclinic. All four deep points give $\hat{k} = 1.30$–1.51 with both intervals excluding one, as does the shallow point at $\beta = 0.18$ ($\hat{k} = 1.33$), where the breathing band has narrowed on the approach to the Takens–Bogdanov point. The three shallow points at $\beta \leq 0.10$, by contrast, are statistically memoryless: $\hat{k} = 0.97$–1.02 with confidence intervals straddling one (at $(\beta, A) = (0.05, 0.43)$ the bootstrap interval $[0.94, 1.00]$ sits marginally below). The capture-time quartiles show why: near the homoclinic they are strongly right-skewed ($9.8/13.5/70.4\,\mathrm{s}$ at $(0.05, 0.43)$ versus $37.5/62.5/85.7\,\mathrm{s}$ at the corner) — a seed either exits promptly or is carried around long ghost-cycle passages near the saddle, and the heavy tail flattens the hazard. Two cautions bound the reading. First, these $\hat{k}$ measure the deterministic reduced flow's dispersal of the seed ensemble, not the finite-N hazard: at the operating corner itself the reduced ensemble gives $\hat{k} = 1.40$ against the measured $k_{\mathrm{abs}} = 2.27$–3.03 (Sec. 4), so the reduced value is a shape test, not a magnitude prediction. Second, and consequently, the rising hazard reported in the body is a property of the operating corner's depth beyond the homoclinic — it holds at that depth for every β tested — but it is not a property of post-homoclinicity per se: within $\approx$0.02 of the homoclinic at $\beta \leq 0.10$ the reduced collapse-time distribution loses its aging shape. This sharpens, in the same direction, the depth caveat of Appendix A.

Table C1 Censored-Weibull shape $\hat{k}$ of the reduced-flow capture times at the eight aging test points ($n = 1200$ canonical-seed initial conditions per point, zero censoring at $t_{\max} = 2000\,\mathrm{s}$). Depth is $A - A_{hc}(\beta)$; σ is the unstable-spiral rate at the chimera fixed point. Profile CI: profile-likelihood 95% interval (Sec. 4 fitter); bootstrap CI: 1,000-resample percentile interval (independent fitter). Bold rows have both intervals above one.

β	A	depth	σ (s^{-1})	$\hat{k}$ [profile CI]	bootstrap CI
0.03	0.461	+0.021	0.0048	1.02 [0.98, 1.07]	[0.98, 1.07]
0.03	**0.500**	+0.060	0.0053	**1.30** [1.24, 1.36]	[1.24, 1.37]
0.05	0.430	+0.020	0.0093	0.97 [0.93, 1.01]	[0.94, 1.00]
0.05	**0.500**	+0.090	0.0124	**1.40** [1.33, 1.46]	[1.33, 1.47]
0.10	0.374	+0.020	0.0126	1.02 [0.98, 1.06]	[0.98, 1.08]
0.10	**0.500**	+0.146	0.0250	**1.51** [1.45, 1.58]	[1.45, 1.59]
0.16	**0.500**	+0.179	0.0398	**1.43** [1.37, 1.49]	[1.39, 1.48]
0.18	**0.339**	+0.020	0.0117	**1.33** [1.29, 1.37]	[1.22, 1.53]

How sharply does the aging shape turn on with depth? Refitting the raw capture times of Table C1 (1,200 per point, zero censoring), the shallow-to-deep contrast at each paired phase lag is $\Delta\hat{k} = +0.28$, $+0.43$, $+0.49$ at $\beta = 0.03$, 0.05, 0.10 (7–13 standard errors), so at every β with paired coverage the trend with depth is monotone; a weighted linear fit to the six $\beta \leq 0.10$ points gives a descriptive slope of 4.6 ± 0.3 per unit depth (capture-time-resampling bootstrap interval $[4.0, 5.0]$), though no two-parameter form fits within the tight per-point intervals. The onset, however, is only bracketed. For $\beta \leq 0.10$ a threshold model ($k = 1$ below a depth d^*, a free level above) is consistent with the three shallow points ($\chi^2 = 4.6$, 3 dof, $p = 0.20$) and places d^* between the deepest memoryless point and the shallowest aging point, $A - A_{hc} \in [0.0205, 0.0595]$; the design has no test point inside this gap, and a 1,000-replicate bootstrap of the raw capture times returns the changepoint to the same gap in every replicate while leaving its position within the gap unconstrained. We therefore report an onset bracket, not an onset depth (Fig. C4a).

Raw depth is, moreover, not the controlling variable across phase lags: the shallow $\beta = 0.18$ point ages ($\hat{k} = 1.33$, $\sim$16 standard errors above one) at the same depth $A - A_{hc} \approx 0.020$ where every $\beta \leq 0.10$ point is memoryless, and no depth threshold separates the eight points (the best two-level split leaves $\chi^2 = 212$). Neither linear-rate covariate explains the exception — this point has the lowest ω of all eight ($0.179\,\mathrm{s}^{-1}$), and its $\sigma = 0.0117\,\mathrm{s}^{-1}$ is indistinguishable from the non-aging $(0.10, 0.374)$ point's $0.0126\,\mathrm{s}^{-1}$ — but the band-narrowing reading given above does, quantitatively: measured in local breathing-band widths, $(A - A_{hc})/(A_{hc} - A_H)$, the $\beta = 0.18$ point lies 1.3 band-widths beyond the homoclinic, deeper than the aging $(0.03, 0.500)$ point at 0.35, and a single threshold in $[0.27, 0.35]$ band-widths separates all eight points — memoryless below, aging above (Fig. C4b). With eight points this collapse is descriptive, not a fitted law; what the data establish is a monotone deepening of the aging shape with depth at fixed β, an onset bracketed in $A - A_{hc} \in [0.0205, 0.0595]$ for $\beta \leq 0.10$, and an onset that moves inward with the width of the breathing band as the Takens–Bogdanov point is approached.

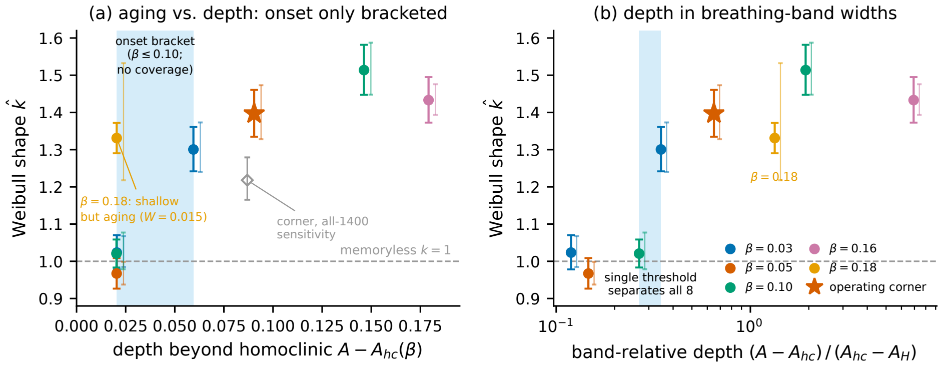

Fig. C4 Depth dependence of the reduced-flow aging shape. (a) Censored-Weibull shape $\hat{k}$ at the eight aging test points of Table C1 versus depth beyond the homoclinic (heavy bars: profile-likelihood 95% intervals; light bars: bootstrap intervals of the independent fitter; star: operating corner; open diamond: the all-1400 corner refit including the degenerate $N = 4$ rows, $\hat{k} = 1.22$ $[1.17, 1.28]$). For $\beta \leq 0.10$ the aging onset lies in the shaded bracket $A - A_{hc} \in [0.0205, 0.0595]$, a coverage gap with no test points; the shallow $\beta = 0.18$ point ages at a depth where every $\beta \leq 0.10$ point is memoryless. (b) The same points against depth measured in local breathing-band widths $(A - A_{hc})/(A_{hc} - A_H)$: a single threshold in $[0.27, 0.35]$ band-widths (shaded) separates all eight points (descriptive, $n = 8$).

Appendix D Ring hazard-shape extrapolation

Section 7 reports that the ring's Weibull shape $\hat{k}$ rises with N over the campaign sizes; this appendix settles whether that rise saturates or diverges as $N \to \infty$. The data are the per-(β, N) censored-Weibull shapes with profile-likelihood intervals over $N \in \{32, 64, 128, 192, 256\}$ (300 runs per cell). Censoring is exactly zero at $N \geq 64$ and at most 9.3% at $N = 32$; at the extrapolation anchors $N = 192$ and 256 the maximum observed lifetime is $\approx$4–6×10^3, far below $T_{\max} = 12{,}000$, so those fits are entirely uncensored.

Per β, we fit $k(N)$ by least squares weighted with the per-point uncertainty σ (the symmetrized profile interval), comparing one bounded form against two divergent ones: the saturating $k(N) = k_\infty - a/N$ (two parameters), the logarithmic $k(N) = b + a \ln N$, and the power law $k(N) = aN^\gamma$. Model selection is by AICc with the small-sample penalty at $n = 5$ points (two-parameter forms cost $\chi^2 + 10$, three-parameter forms $\chi^2 + 30$), and we call the comparison decisive when the AICc gap between the best divergent and best saturating form is at least 2. Uncertainty on k_∞ is bootstrapped: 5,000 Gaussian resamples of the $\hat{k}$ values within their σ, refit, percentile intervals.

The saturating form wins in all five β cells, with gaps 2.29–5.42 (Table D2). Adding $N = 192$ — the firming-up step that made a three-parameter fit AICc-valid — widened the three previously marginal gaps ($2.32 \to 2.98$ at $\beta = 0.110$; $2.10 \to 2.29$ at $\beta = 0.120$; $3.87 \to 4.65$ at $\beta = 0.130$) and weakened none. The general three-parameter form $k(N) = k_\infty - aN^{-\gamma}$ serves as confirmation rather than selection (its third parameter costs +20 AICc at $n = 5$): its k_∞ agrees with the bounded fit to within 0.02–0.04 everywhere, and its $\gamma \approx 1.1$–$1.6 > 1$ says the saturation is at least as fast as $1/N$, so the two-parameter form's fixed $\gamma = 1$ is a slightly conservative

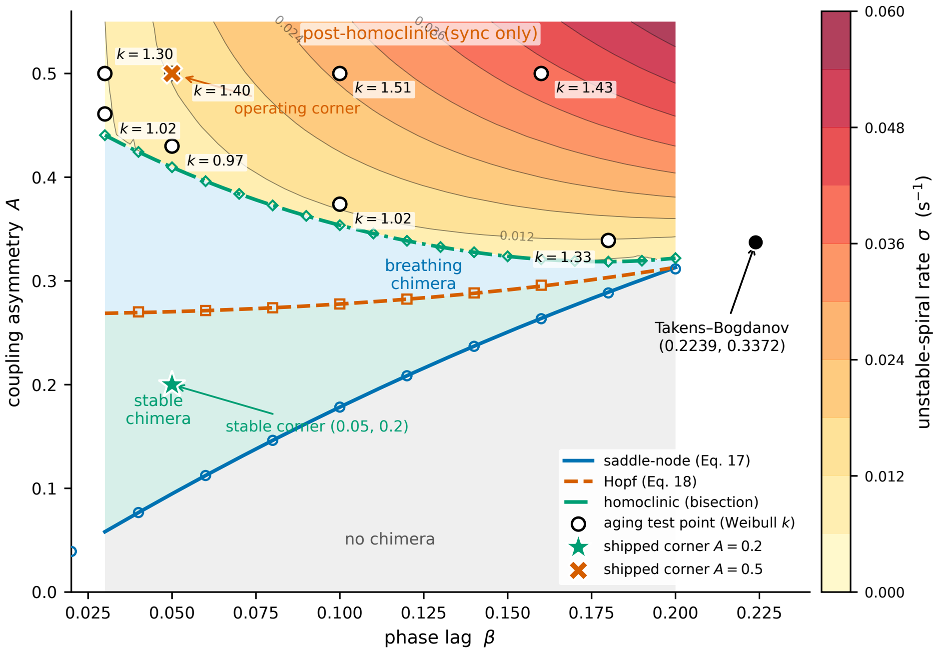

Fig. C5 Corner-generality map. The four regimes of the (β, A) plane bounded by the saddle-node (Eq. 17), Hopf (Eq. 18) and homoclinic curves (the latter traced at 35 values of β, bisection bracket $\leq 7 \times 10^{-5}$), meeting at the Takens–Bogdanov point $(0.2239, 0.3372)$. The post-homoclinic wedge is shaded by the unstable-spiral rate $\sigma(\beta, A)$ of the persisting chimera fixed point ($\sigma > 0$ at all 592 grid points; labelled contours in s^{-1}). Open circles: the eight aging test points with their fitted censored-Weibull shape $\hat{k}$ (Table C1); $\hat{k} > 1$ at all four deep points and at the shallow $\beta = 0.18$ point, while the shallow points at $\beta \leq 0.10$ (depth $\approx +0.02$) are consistent with a memoryless collapse. Star and cross: the two shipped corners.

choice. The nominally divergent fits barely diverge in any case: the fitted logarithmic slopes are 0.05–0.18 and the power exponents 0.04–0.13. All five bootstrap k_∞ intervals exclude one and are tight (width ≤ 0.18), and k_∞ rises monotonically toward the existence boundary: $1.35 < 1.49 < 1.56 < 1.62 < 1.65$ for $\beta = 0.110 \rightarrow 0.130$. The boundary itself sits above the sweep: Appendix E measures $\beta_c = 0.137$ $[0.130, 0.144]$, so the entire sweep approaches β_c from below and the rise of k_∞ is a rise *toward* the boundary, not across it (bootstrap probability 0.966 that β_c exceeds the last sweep point 0.130).

The Sec. 7 validations carry over to the extrapolation anchors. At $N = 256$ the dying object is a canonical chimera in 97–98% of runs (equilibration-gated classification of the re-run fields), and the equilibration-filtered refits — dropping runs in which no chimera ever established together with the rare degenerate channel — leave $\hat{k}$ unchanged or slightly larger, intervals still excluding one. The hazard shape is criterion-robust across the spatial-homogeneity family ($\rho_{\text{std}} < 0.04, 0.08, 0.10$, all with 0% censoring), and the one criterion that appears to disagree — the mean-coherence level rule — is the diagnosed twisted-state censoring artifact of Sec. 7 (Fig. D6).

Table D2 Saturate-versus-diverge model selection for the ring's $\hat{k}(N)$, per β. k_∞ is the bounded-fit asymptote with bootstrap 95% interval; the gap is AICc(best divergent) − AICc(best saturating), decisive at ≥ 2; γ is the exponent of the confirmatory three-parameter saturating fit.

β	$\hat{k}(N = 32 \ldots 256)$	k_∞ [95% CI]	AICc gap	γ
0.110	$1.13, 1.34, 1.35, 1.38, 1.22$	1.35 $[1.27, 1.43]$	$+2.98$	1.63
0.115	$1.15, 1.52, 1.42, 1.43, 1.37$	1.49 $[1.40, 1.56]$	$+5.42$	1.51
0.120	$1.15, 1.36, 1.49, 1.46, 1.53$	1.56 $[1.48, 1.65]$	$+2.29$	1.14
0.125	$1.26, 1.64, 1.42, 1.58, 1.60$	1.62 $[1.53, 1.71]$	$+2.51$	1.48
0.130	$1.33, 1.55, 1.68, 1.68, 1.47$	1.65 $[1.56, 1.74]$	$+4.65$	1.51

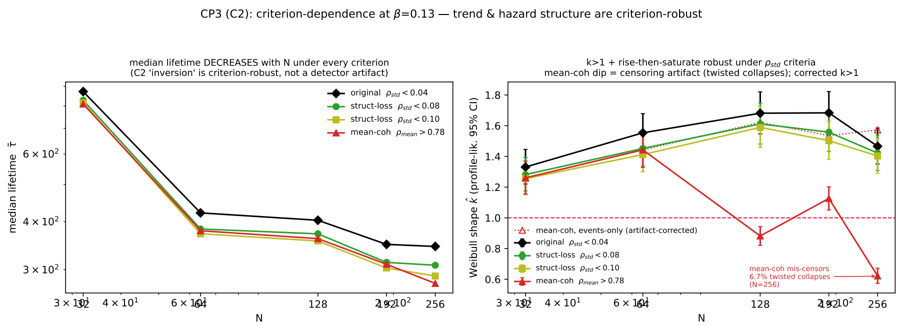

Fig. D6 Criterion-independence at $\beta = 0.130$. Left: median lifetime decreases with N under every death criterion (the near-boundary inversion is not a detector artifact). Right: the Weibull shape $\hat{k}(N)$ stays above one under the spatial-homogeneity (ρ_{std}) criteria; the mean-coherence rule's apparent dip below one at large N is a censoring artifact (twisted-state collapses mislabeled as censored), and the artifact-corrected curve returns above one.

Two caveats bound the reading without weakening the call. First, four of the five cells are non-monotonic in $\hat{k}(N)$ (in-band scatter about the plateau; weighted $\chi^2 = 8$–11 on 3 residual degrees of freedom), with only $\beta = 0.120$ a clean leveling fit ($\chi^2 = 0.70$): k_∞ should be read as the plateau *level*, ≈1.35–1.68, not a precisely pinned asymptote. Second, the three-parameter fit is partially degenerate at five points (a and γ are not separately identified), which is why the headline rests on the parsimonious two-parameter form. What is firm is the binary outcome and its direction: the saturating form wins in 5/5 cells (0/5 divergent), and the limiting shape exceeds one in every cell — the increasing-hazard result survives the thermodynamic limit.

Appendix E An independent measurement of the ring existence boundary

The campaign sections take the ring chimera's existence boundary $\beta_c \approx 0.13$–0.14 from the literature; this appendix turns that citation into a measurement with the same frozen pipeline (integrator, initial conditions, death criterion $\rho_{\text{std}} < 0.04$ sustained 50 time units, $T_{\max} = 12{,}000$). The operational endpoint was fixed before the ladder ran, on design probes only. Per (β, N) cell, P_{persist} is the fraction of runs that (a) establish a chimera — the equilibration gate of Sec. 7, maximum ρ_{std} over the live window $[50, \tau]$ exceeding 0.08 — and (b) survive past a fixed horizon $T^* = 400$ time units, the pooled median lifetime of the last pre-registered cell $\beta = 0.130$ (medians 422/404/345 at $N = 64/128/256$), roughly twice the ≈200–250 decay time of the initial-condition transient that survives above the boundary. The boundary estimate $\beta_c(N)$ is the midpoint of a four-parameter logistic (free floor and ceiling, binomial maximum likelihood) fit to $P_{\mathrm{persist}}(\beta)$ over the ladder $\beta = 0.115$–0.170 in steps of 0.0025 (100 runs per cell, $N \in \{64, 128, 256\}$, 6,900 runs, zero censoring, globally unique seeds disjoint from all prior campaigns); intervals are percentile bootstrap over runs within cells ($B = 2{,}000$).

Establishment never fails appreciably ($\geq 95\%$ in every cell): crossing the boundary does not prevent the seeded chimera from forming, it shortens its life. P_{persist} falls from a plateau of 0.59–0.75 at $\beta \leq 0.1225$ to a floor of 0.05–0.29 at $\beta \geq 0.16$ (Fig. E7a) — the floor is nonzero because slowly decaying traveling-wave remnants occasionally outlive T^*. The logistic midpoints are $\beta_c(64) = 0.130$ $[0.128, 0.136]$, $\beta_c(128) = 0.136$ $[0.128, 0.140]$, and $\beta_c(256) = 0.134$ $[0.129, 0.141]$. Extrapolating $\beta_c(N) = \beta_c^\infty + a/N$ (weighted least squares; the $1/N$ form beats $1/\sqrt{N}$ on χ^2, 0.68 versus 0.87 on one degree of freedom) gives

$$\beta_c^\infty = 0.137\ [0.130, 0.144],$$

with the $1/\sqrt{N}$ form at 0.141 $[0.128, 0.152]$ — a form systematic of ≈0.004, smaller than the statistical interval. Moving the horizon to $T^* = 300$ or 500 moves the extrapolated midpoint to 0.142 or 0.135, a comparable systematic; under every variant the boundary stays inside 0.13–0.15. The robustness endpoint — the knee where the median lifetime of established runs meets the post-boundary transient floor (Fig. E7b) — sits higher, at $\beta = 0.155$–0.164 across N: the persistence midpoint marks the center of the crossover, the knee its upper edge, and together they say the boundary is a soft crossover in this finite-N, fixed-criterion setting rather than a sharp threshold. As a consistency check, the two ladder rungs shared with the main campaign ($\beta = 0.125, 0.130$; disjoint seeds, 100 versus 300 runs) reproduce the published median lifetimes within sampling error.

Two readings follow. First, the measured $\beta_c^\infty = 0.137$ $[0.130, 0.144]$ lands inside the literature range 0.13–0.14 the manuscript previously only cited. Second, the k_∞ ladder of Appendix D ($1.35 \to 1.65$ over $\beta = 0.110 \to 0.130$) rises toward a boundary that sits above its last rung with bootstrap probability 0.966 (Fig. E7d), so "rising monotonically toward β_c" is geometrically coherent — with the honest caveat that the bootstrap interval grazes 0.130 rather than excluding it outright, and that $\beta_c(N)$ is measured at three sizes, so the extrapolation leans on the form-robustness reported above.

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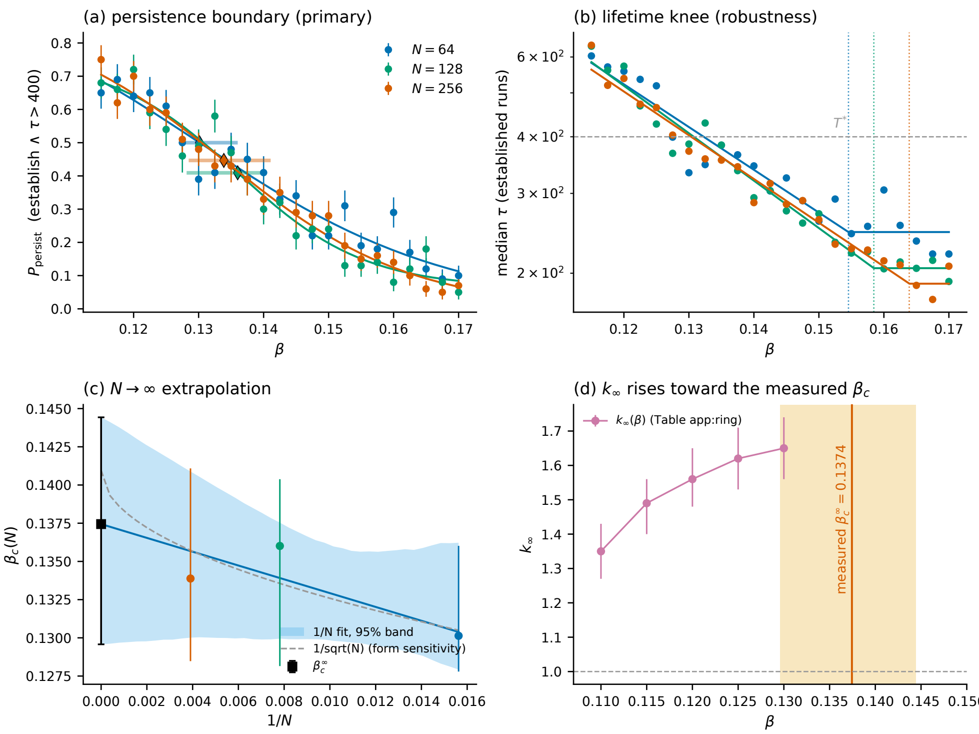

Fig. E7 Independent measurement of the ring existence boundary. (a) $P_{\text{persist}}(\beta)$ (establish $\wedge$ $\tau > T^* = 400$) per N with four-parameter logistic fits; diamonds mark the midpoints $\beta_c(N)$ with bootstrap 95% intervals. (b) Median lifetime of established runs: a decline meeting the post-boundary transient floor at the knee (dotted verticals), the robustness endpoint. (c) $\beta_c(N)$ versus $1/N$ with the bootstrap band of the $1/N$ extrapolation and the $1/\sqrt{N}$ form for the systematic; the square is $\beta_c^\infty = 0.137$ $[0.130, 0.144]$. (d) The $k_\infty(\beta)$ ladder of Table D2 rises toward the measured boundary (vertical line; shaded 95% interval).

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